



Obstetric Ultrasound: Dating, Imaging, and Growth

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KEY ABBREVIATIONS

Abdominal circumference	AC	Food and Drug Administration	FDA
American College of Obstetricians and Gynecologists	ACOG	Head circumference	HC
American Institute of Ultrasound in Medicine	AIUM	Kilohertz (1000 cycles per second)	kHz
Amniotic fluid index	AFI	Last menstrual period	LMP
As low as reasonably achievable	ALARA	Megahertz (1 million cycles per second)	MHz
Biparietal diameter	BPD	National Institute of Child Health and Human Development	NICHD
Crown-rump length	CRL	Small for gestational age	SGA
Current Procedural Terminology	CPT	Society for Maternal-Fetal Medicine	SMFM
Estimated date of delivery	EDD	Spatial—peak temporal average	SPTA
Estimated fetal weight	EFW	Three-dimensional	3D
Femur length	FL		
Fetal growth restriction	FGR		

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OVERVIEW

Ultrasound imaging is undoubtedly the most valuable diagnostic tool in obstetrics. Since the first published report of its use in pregnancy in 1958,¹ ultrasound has become widely used in various aspects of obstetric care, including the evaluation of the placenta, fetal biometry, and the diagnosis of fetal anomalies. Obstetric ultrasound resolution and portability has continued to evolve, with ultrasound technology now present in almost all obstetrics offices and labor and delivery units. This chapter aims to address the general aspects of ultrasound in pregnancy; the use of ultrasound to detect fetal anomalies is addressed in Chapter 10. More detailed discussions of other specific pregnancy problems that include ultrasound assessment—such as multiple gestation, third-trimester bleeding, placenta accreta, and cervical insufficiency—are covered in other chapters.

BIOPHYSICS OF ULTRASOUND

The underlying basis of ultrasound image production relies on the piezoelectric effect: when electrical impulses are applied to certain ceramic crystals, mechanical oscillations are induced. Conversely, induced vibrations of piezoelectric crystals generate a detectable electric current. In diagnostic ultrasound applications, the ultrasound machine sends an electrical signal of the desired frequency to piezoelectric crystals embedded in the ultrasound probe. When the probe is placed in contact with a patient's skin, the skin and underlying tissues begin to vibrate, generating a sound or pressure wave. As this pulse of energy encounters an interface between materials of different impedance, a small amount of the energy is reflected as an echo. The pulses that return to the patient's skin cause the crystals in the probe to vibrate, which generates an electric current that is passed back to the ultrasound machine. Based on the intensity, timing, and location of the returning signals, the ultrasound machine produces the image that is displayed on the monitor. The pulses of energy emitted with diagnostic ultrasound are very brief—about 1 μ s. The number of pressure peaks produced in 1 second approximates the frequency of sound waves. Ultrasound machines used in obstetrics operate at frequencies between about 2 and 9 MHz. Sound frequencies above 20 kHz cannot be detected by the human ear, hence the term *ultrasound*.

SAFETY OF ULTRASOUND

Fetal Safety

Since diagnostic ultrasound was first introduced for clinical use, attention has appropriately been given to ensuring its safety. Ultrasound in high intensities has the potential to adversely affect fetal tissue in two ways, both of which are mitigated by user monitoring. The first way is “mechanical effect.” Ultrasound micropulses are sent out in such a way that the time between each pulse allows the transducer to appreciate returning echoes so that the transducer spends about 1% of the time emitting and 99% of the time “listening.” The mechanical effect, which can be monitored by the mechanical index (MI), is based on the peak pulse intensities (PPI) and at high levels (well above usual diagnostic intensities) can cause cell damage through cavitation, a phenomenon wherein agitation of sufficient strength can adversely affect the integrity of cells. This is the principle behind lithotripsy, which uses low-frequency ultrasound at very high peak focused intensities. Fortunately, this can only occur when peak pulse

intensities far exceed those used diagnostically. However, as an additional safety factor, manufacturers are required to apprise the user of the intensities being employed by displaying them on the ultrasound screen.¹

The second type of potential risk of ultrasound is a thermal effect. The thermal effect refers to the possibility that energy from sound waves is converted to heat as sound waves are attenuated during the passage through tissue. This presents potential safety concerns, as it is well recognized that an increase in temperature above 1.5°C from the physiologic level can be harmful to developing embryos.² The thermal effect is not based on the strength of acoustic pulses but on an average summation of exposure over time—the spatial peak temporal average (SPTA). Again, this is displayed on the ultrasound screen as a thermal index (TI) either in bone or soft tissue. This is mostly problematic when pulsed wave Doppler ultrasound is used since longer pulses are needed to appreciate the change in Doppler frequency occurring when an ultrasound beam encounters blood flow traveling toward, or away from, a transducer. This results in shorter time intervals between pulses and, therefore, these resulting changes to the duty cycle (time off versus time on) will increase the (averaged) SPTA. The user can monitor the potential thermal effect on tissue by following the TI on the screen. The likelihood of attaining an SPTA high enough to interfere with tissue integrity with TIs below 1.0 is negligible.³ A 2015 official statement from the American Institute of Ultrasound in Medicine (AIUM) on ultrasound bioeffects states that no independently confirmed effects have been found from unfocused beam SPTA intensities below 100 mW/cm² or TI values of less than 2.⁴

In 2009, the World Health Organization sponsored a systematic review and meta-analysis to evaluate the safety of human exposure to ultrasonography in pregnancy. This study showed no adverse pregnancy outcomes, impaired physical or neurologic development, increased risk for malignancy in childhood, subnormal intellectual performance, or mental diseases.⁵ This analysis concluded that according to available evidence, exposure to diagnostic ultrasonography during pregnancy appears to be safe. Nevertheless, the American College of Obstetricians and Gynecologists (ACOG), the AIUM, and the International Society of Ultrasound in Obstetrics and Gynecology (ISUOG) all caution that the possibility of harm exists, and maintain that ultrasound should be used only by qualified health professionals to provide medical benefit to the patient.⁶⁻⁸ As is the case with other uses of diagnostic ultrasound, the use of ultrasound in pregnancy should adhere to the ALARA principle (as low as reasonably achievable). This means that the potential benefits and risks of each examination should be considered, and equipment controls should be adjusted to reduce as much as possible the acoustic output from the transducer. It is imperative that ultrasound operators are familiar with this concept.

Preventing Transmission of Infections

To avoid transmitting disease from one patient to another, proper cleansing and disinfecting of ultrasound probes is important. The relevance of this was highlighted in the height of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic.^{8a} With a transabdominal probe, an appropriate disinfectant wipe should be used in accordance with local regulations. Transvaginal probes should be covered during use with a disposable latex or nonlatex cover. Following the exam, the probe should be cleaned with running water or a damp cloth. It is then important that chemical high-level disinfection be carried out as per the probe manufacturer's recommendations.

WHAT MAKES A GOOD ULTRASOUND STUDY?

Proper Gain and Focus

“Gain” refers to the lightness or darkness of ultrasound images. Acceptable images demonstrate the full range of grays that the machine is capable of producing. Gain is controlled by simply adjusting a knob found on all ultrasound machines. With improper gain, important detail is lost (eFig. 9.1). Electronic control of the timing and order in which crystals are activated can work to focus this beam at the region of interest within tissues. Image resolution is optimal when the structure of interest lies within this zone of optimal focus, which can be adjusted by the sonographer.

Optimal Magnification

Proper depth of the scanned field and magnification of areas of interest are important for several reasons. Limiting the size of the scanned area allows a higher frame rate and resolution. In addition, homing in on the area of interest draws attention to important details within that scanned area (eFigs. 9.2 and 9.3).

Attenuation and Shadowing

Propagation of ultrasound waves is affected by the medium through which the sound waves pass. For several reasons, ultrasound waves lose intensity as they pass through tissues. The pressure waves gradually diverge from the central beam and are scattered by reflection from small structures within the tissue, and part of the sound energy is absorbed within tissues. Some tissues, such as bone, strongly attenuate sound waves (eFig. 9.4). The thicker the tissues through which sound waves must pass before arriving at the target, the more the attenuation and the greater the difficulty in retrieving useful information from the echoes. Because of attenuation, obstetric ultrasound imaging is greatly affected by patient obesity. The use of “windows” in the abdominal wall of the pregnant patient around the umbilicus and below the pannus can greatly enhance image quality (eFig. 9.5). In early pregnancy or when structures of interest are low in the pelvis, the problem of attenuation is reduced considerably by the use of an endovaginal probe. When transvaginal ultrasound is used, a full bladder is usually not necessary.

Lower-frequency sound waves penetrate tissues better but cannot achieve the same resolution as higher-frequency oscillations. Available ultrasound probes operate over a range of frequencies (e.g., ranges of 1 to 5 MHz and 4 to 8 MHz for abdominal probes). The highest-frequency probe that allows adequate penetration should be used. Because penetration of the abdominal wall is not an issue with endovaginal ultrasound, probes used for this purpose usually operate at a frequency of 5 to 10 MHz. Regardless of the habitus, we recommend using the highest-frequency probe that is available for the initial approach and then moving down in frequency as needed to aid for penetration.

Power Adjustment

Power, the energy that is emitted from the ultrasound, is different from gain, which is the degree of amplification of returning echoes. Increasing the power setting can improve penetration; however, this has much less effect on attenuation and optimal image production than does use of the proper probe frequency or natural windows.

Proper Orientation

By convention, in sagittal views of the uterus of the pregnant patient, the right of the screen corresponds to the inferior aspect of the patient (Fig. 9.1). For transverse views, the patient's right is shown on the left

of the screen. With transvaginal ultrasound, transverse views also show the patient's right side to the left of the screen. In transvaginal sagittal views, *up* is to the left of the screen (i.e., toward the pregnant patient's bladder), and *down* (toward the pregnant patient's sacrum) is to the right of the screen (Fig. 9.2). Ultrasound transducers have a notch or ridge that demarcates the side of the probe that will correspond to the left side of the monitor. Thus in going from sagittal to transverse scans, the probe is always rotated counterclockwise, and to move from transverse to sagittal, it is rotated clockwise.

A *backward* image where the patient's “left” is “right” and “up” is “down” on the screen is typically considered to be unacceptable for diagnostics and for documentation because of the risk of misinterpretation. However, certain providers might intentionally choose to



Fig. 9.1 Transabdominal Sagittal View of the Uterus. The uterine fundus and cervix are labeled. By convention, the right side of the ultrasound screen corresponds to the inferior aspect of the patient.

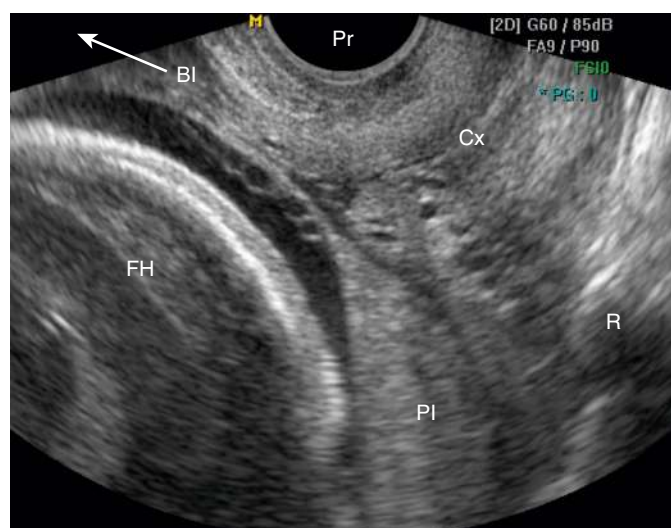
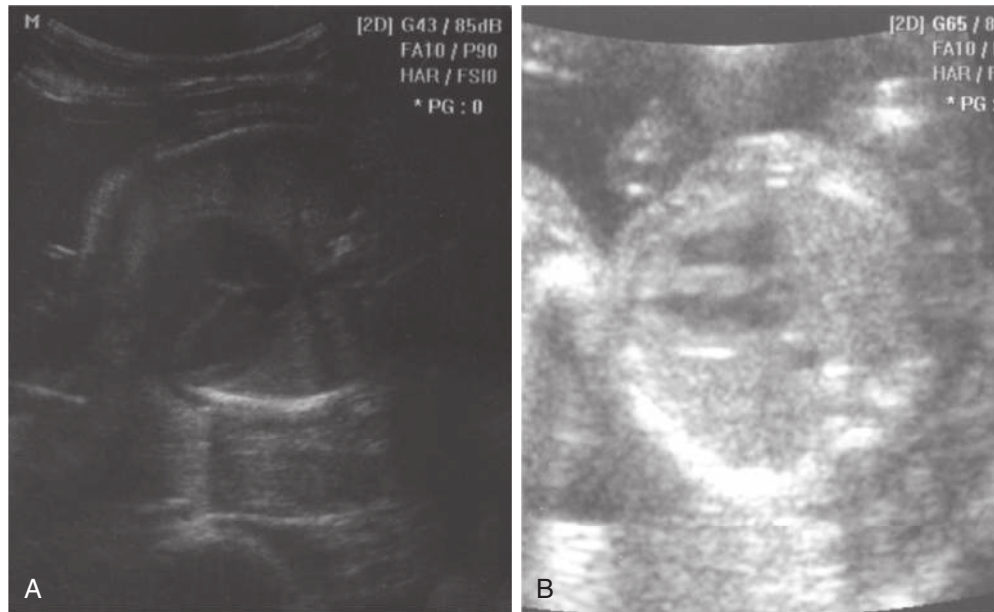
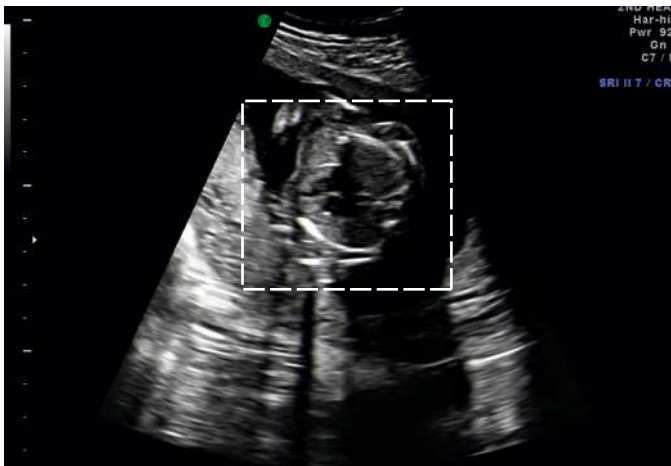


Fig. 9.2 Transvaginal Sagittal View Showing Proper Orientation. The left of the screen is “up” on the patient, that is, toward the bladder (*Bl*, arrow). Note how transvaginal ultrasound provides the ultimate window, showing very clear views of structures close to the vaginal apex. Also note the presence of vasa previa. *Cx*, Cervix; *FH*, fetal head; *PI*, placenta; *Pr*, probe tip; *R*, rectum.



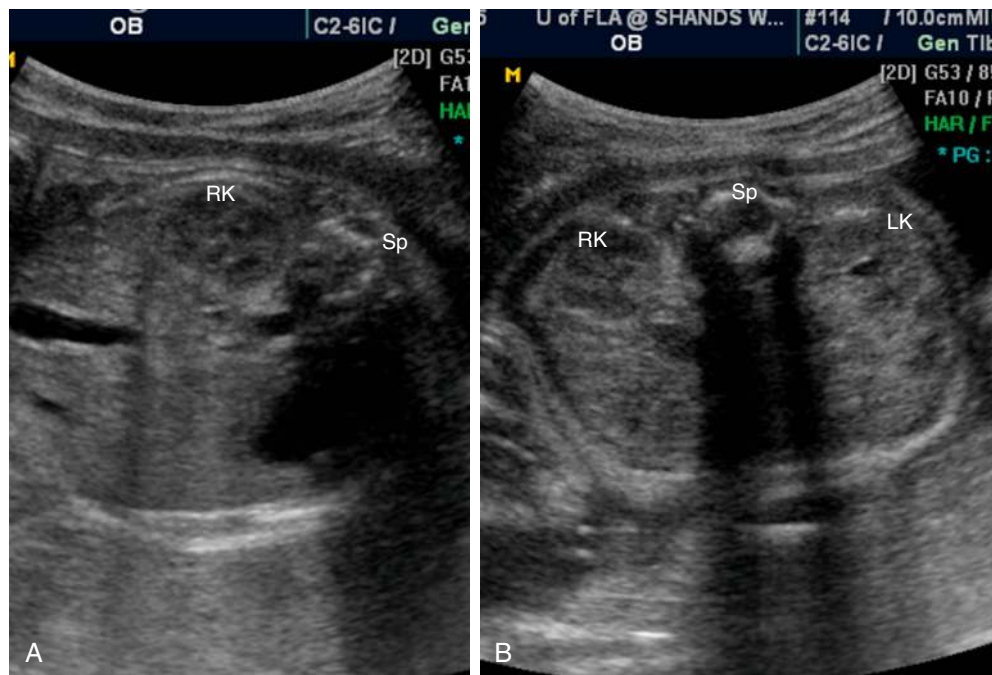
eFig. 9.1 Loss of Detail in Ultrasound Image With Improper Gain. (A) Overall gain is too low, yielding a dark image. (B) Too much gain. In both cases, diagnostic detail is lost.



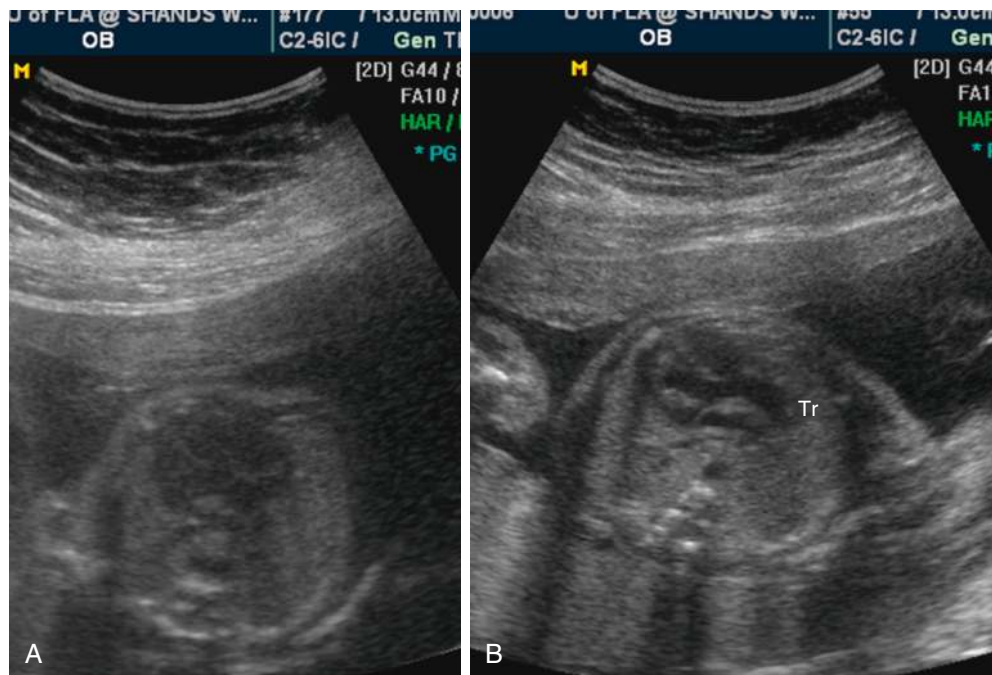
eFig. 9.2 Ultrasound Image Before Zoom Is Applied. The area within the *dashed lines* should be expanded to fill the entire screen.



eFig. 9.3 Appearance After Zoom Is Applied. Compared with eFig. 9.2, the cardiac structures are more easily seen. Additionally, because a smaller area was being scanned, the frame rate increased from 25 to 58 frames/s.



eFig. 9.4 (A) Shadowing by the spine precludes visualization of the left kidney. (B) By sliding the transducer to a different location on the maternal abdominal wall, a more favorable angle of insonation is possible. *LK*, Left kidney; *RK*, right kidney; *Sp*, spine.



eFig. 9.5 (A) Scan through the pannus of the abdominal wall. (B) By moving the probe into the relatively thinner area near the maternal umbilicus, the resolution improved dramatically. A similar result can be obtained when scanning below the pannus in the suprapubic area. This fetus has truncus arteriosus (*Tr*).

hold their probe in a different orientation during an invasive diagnostic procedure such as amniocentesis. If these images are saved, proper labeling is recommended to avoid future misinterpretation. To establish the position of the fetus, the orientation of the probe must be correct. The sonographer should not depend on the side of the stomach or axis of the heart to define the position of the fetus because these structures are not always in a normal position (e.g., heterotaxy syndromes).

Angle of Insonation

There is often a best angle from which to view aspects of the intra-uterine contents and fetal structures. For example, transverse views of the kidneys are best obtained when the fetus is scanned through its back or front. When the angle of insonation is from the side of the fetus, one kidney is shadowed by the spine (see eFig. 9.4). As will be noted later, standard measurements of fetal urinary tract dilation require that the angle of insonation is through the fetal front or back. Optimal views of the brain, heart, and spine also require that the angle of insonation be appropriate. Sometimes, significant pressure with the transducer is needed to get the probe into position for optimal visualization. Additionally, to achieve this ideal angle an ultrasound operator should not hesitate to reposition a pregnant person as needed.

SPECIAL ULTRASOUND MODALITIES

M-Mode

For most obstetric applications, the familiar two-dimensional (2D) gray-scale real-time ultrasound is used. This is formally known as *B-mode ultrasound*. Another ultrasound modality that is available on most machines is referred to as *M-mode ultrasound* (motion mode). M-mode ultrasound shows changes along a single ultrasound beam over time. M-mode is useful for documenting the presence and rate of fetal cardiac contractions (Fig. 9.3, eFig. 9.6) and is also used for specialized echocardiography applications, such as defining cardiac arrhythmias (eFig. 9.7).

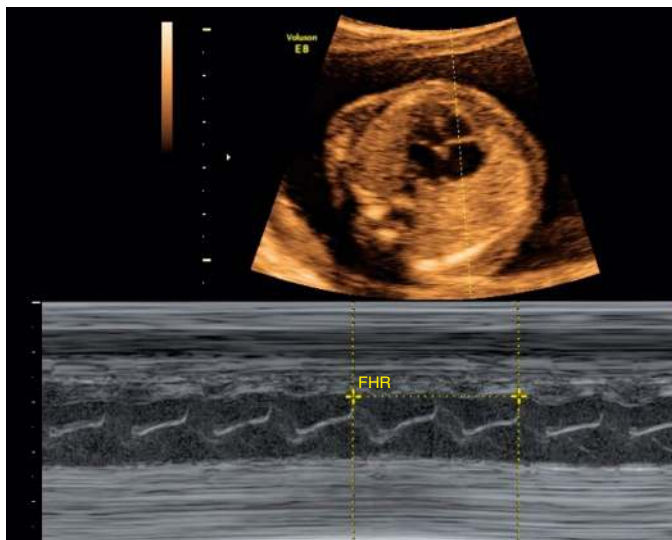


Fig. 9.3 M-Mode Ultrasound with the Cursor Through the Heart Valves. In any ultrasound exam, it is important to have a permanent record showing fetal viability. M-mode is convenient for this purpose and can be used to determine the fetal heart rate (FHR).

Color and Pulsed Wave Doppler

Doppler ultrasound is used to demonstrate the presence, direction, and velocity of blood flow. The machine displays moving blood as color superimposed on the 2D gray-scale image. By convention, flow toward the ultrasound transducer is displayed in red and flow away is displayed in blue, which is also shown in the color scale located in the left margin of the image (Fig. 9.4). Pulsed wave Doppler continuously measures the relative velocity of flow within a designated gate inside a vessel. Flow velocity waveforms are used to calculate the systolic/diastolic (S/D) ratio, the pulsatility index, and the resistance index. These indices are primarily used to assess downstream resistance in the vessel being interrogated. In pregnancies with fetal growth restriction (FGR), the flow within the umbilical artery is used to assess placental function (Fig. 9.5); this is further discussed in Chapter 30.



Fig. 9.4 Color and Spectral Doppler Evaluation of the Umbilical Artery. Coiling arteries and vein are shown at left. Red indicates flow toward the transducer and blue is flow away. The sample gate for the pulsed wave Doppler is superimposed. Normal flow velocity waveform for the umbilical artery is shown at right.

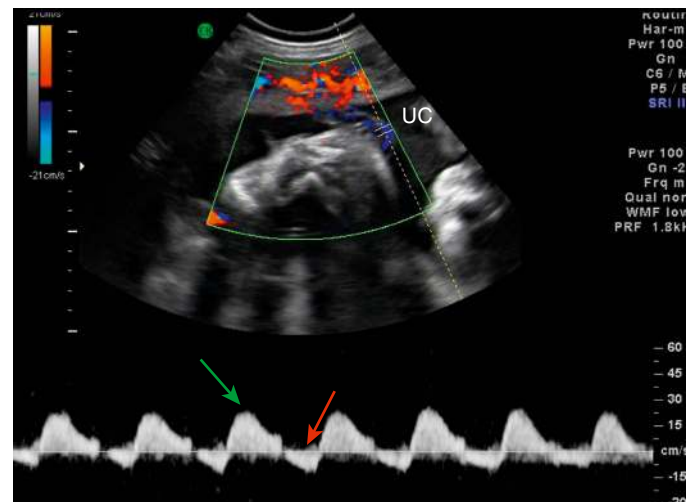
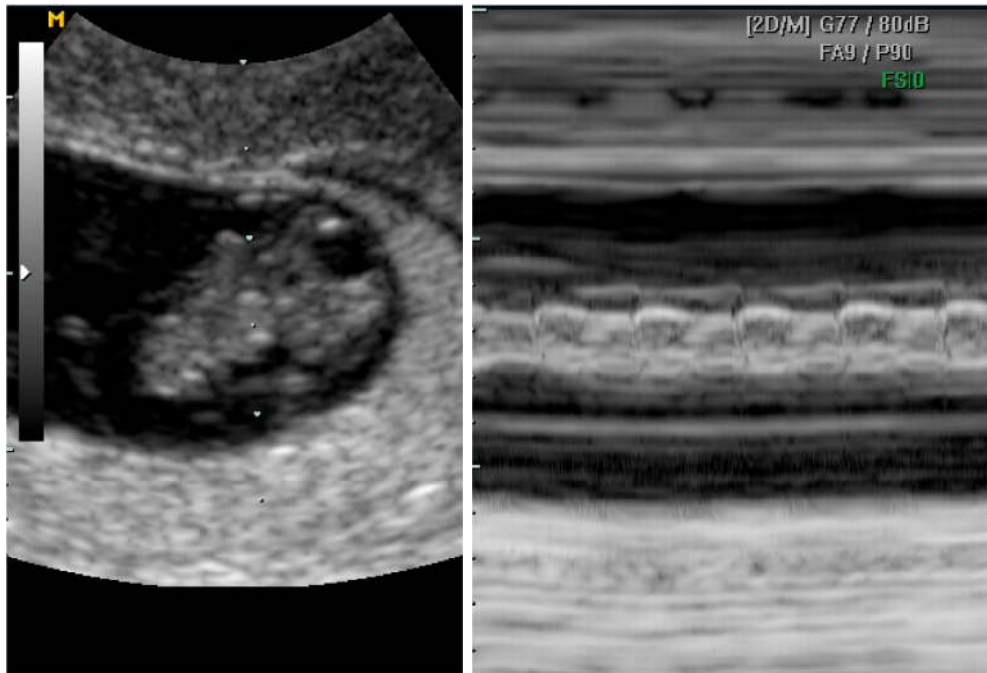
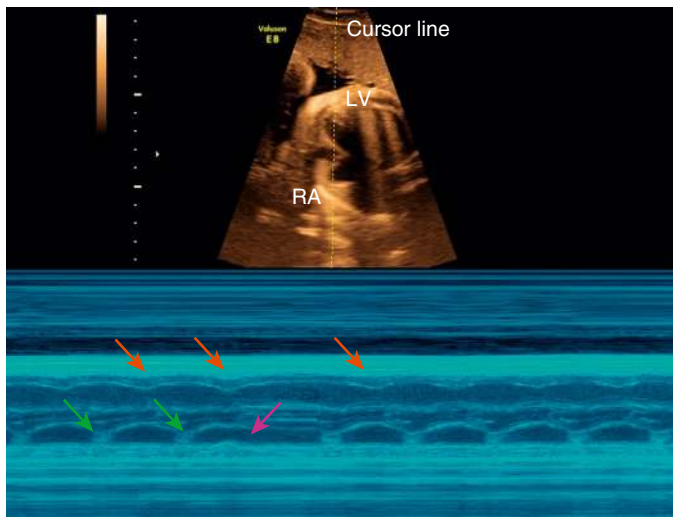


Fig. 9.5 Pulsed Wave Doppler of the Umbilical Cord (UC) Showing Reverse End-Diastolic Velocity. There is forward flow during systole (green arrow) and reverse flow during diastole (red arrow). This indicates very high placental resistance.



eFig. 9.6 M-Mode Application in an 8-Week-Old Fetus. The row of dots in the *left panel* indicates the location of echoes, in this case cardiac pulsations, being displayed over time in the *right panel*. M-mode is preferred to Doppler for documenting fetal viability before 10 weeks. Note the prominent brain vesicle in the fetal head, a normal finding.



eFig. 9.7 M-Mode Ultrasound Demonstrating Premature Atrial Contractions. The cursor line passes through the left ventricle (LV), then the right atrium (RA), in the upper display. The movement along the cursor line over time is shown in the lower display. The *red arrows* show ventricular contractions, and the *green arrows* show atrial contractions. Note that when there is a premature atrial contraction (*magenta arrow*), the ventricle is in a refractory state, resulting in a pause between contractions.

For some applications, the absolute flow velocity is needed. For example, when screening for fetal anemia, the peak flow velocity in the fetal middle cerebral artery is measured, as this correlates with the degree of fetal anemia. To give meaningful results for anemia assessment, the angle of insonation (θ) should be in line with the direction of blood flow (eFig. 9.8). Typically, an angle of 30 degrees or less is recommended for Doppler studies.

Three-Dimensional Ultrasound

Three-dimensional (3D) ultrasound is a volumetric imaging technology that allows for the acquisition of a region of interest in a volume dataset, as opposed to a single slice in the conventional 2D ultrasound. Several techniques can be used to obtain 3D ultrasound volumes. The first technique is a freehand technique that relies on the operator manually moving about the region of interest; a second technique uses an internal mechanical sweep mechanism that automatically obtains multiple 2D images in the region of interest.⁹ The volume data acquired through this process can then be visualized, analyzed, or processed in various ways. Multiplanar mode allows for reconstruction of orthogonal planes from the initial acquisition plane. An example of this would be the reconstruction of a midsagittal plane from an axial acquisition plane (Fig. 9.6). Surface rendering mode can be used to visualize surface structures such as facial features. This modality is often useful in counseling due to its lifelike appearance (Fig. 9.7). Spatial and temporal image correlation (STIC) is a modality in 3D ultrasound that provides an acquisition of a fetal heart volume over a few seconds. 3D ultrasound can also be used to estimate volumes of various structures, such as fetal lung or thigh volumes (eFig. 9.9), with modalities such as virtual organ computer-aided analysis (VOCAL) software (VOCAL Technologies, Ltd., Buffalo, NY).

Despite the advances of 3D ultrasound, its advantages vary from case to case. The use of 3D ultrasound is helpful in visualizing various fetal brain structures that can be challenging to detect, such as the corpus callosum and cerebellar vermis.¹⁰ However, its added value in



Fig. 9.7 Surface Rendering of Fetus with Bilateral Cleft Lip.

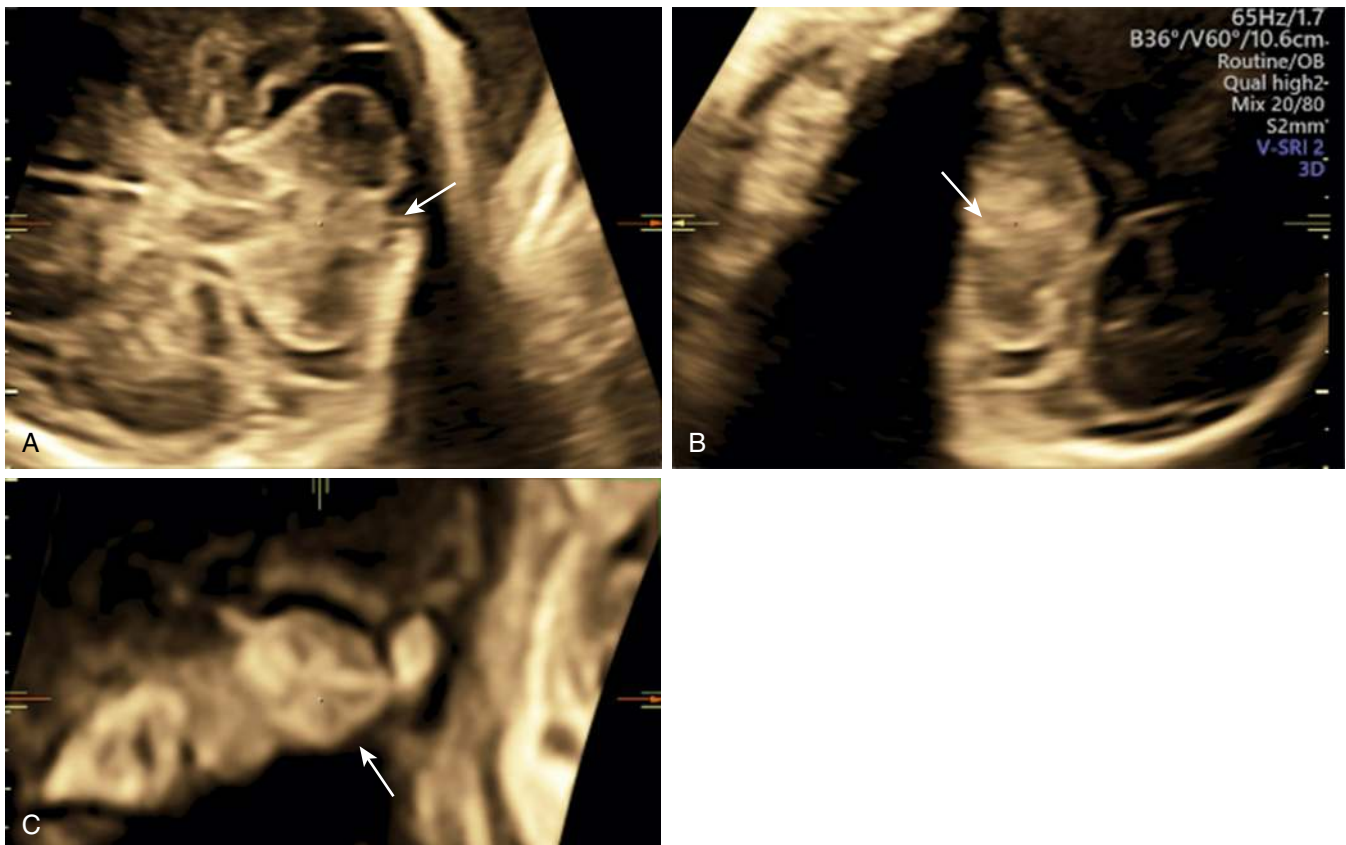
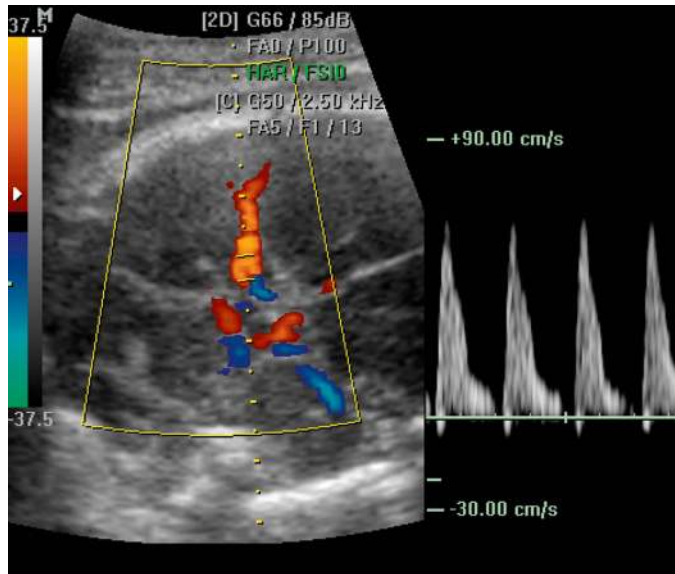
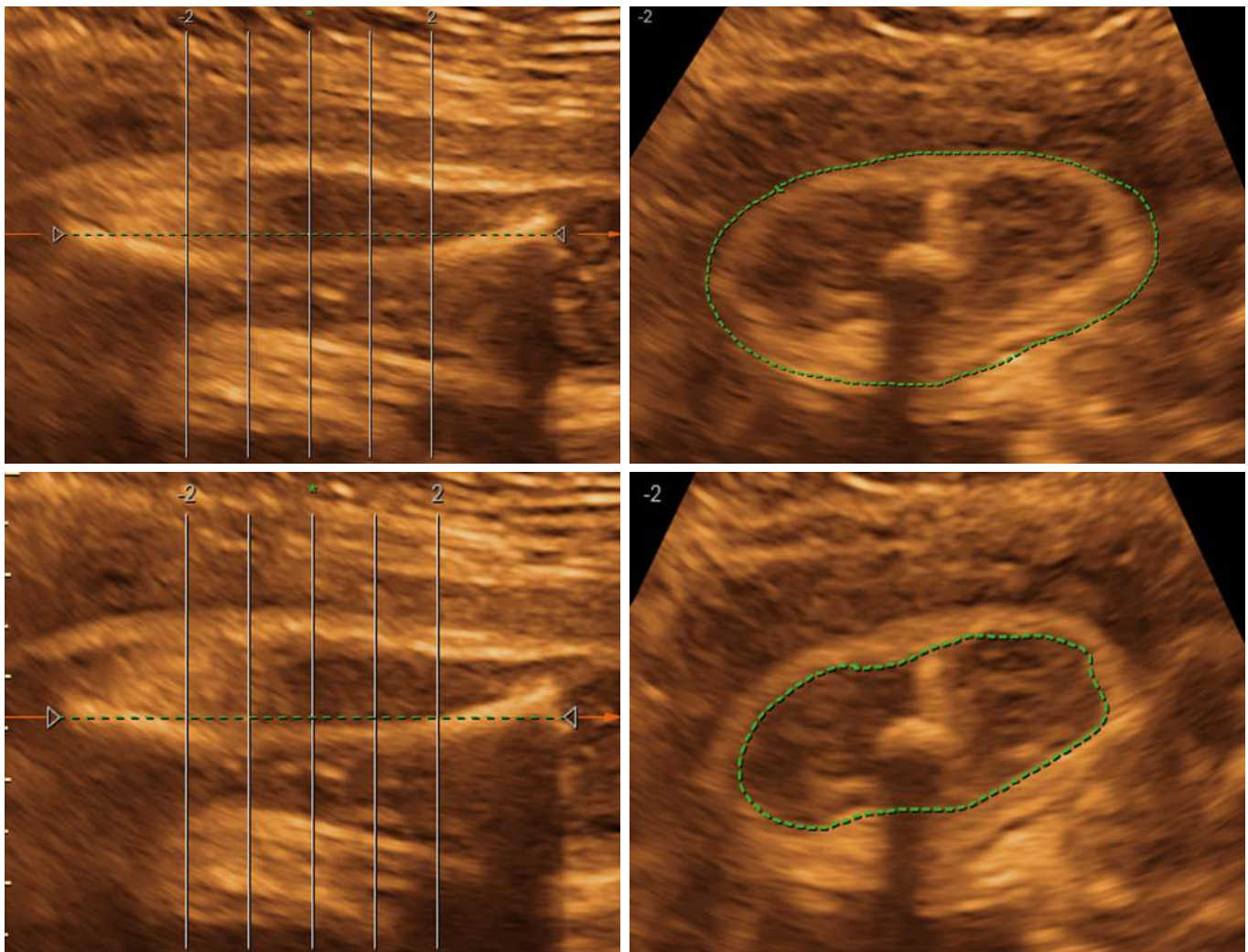


Fig. 9.6 Multiplanar View of the Fetal Vermis. Using axial view as the starting plane (A), acquisition of the sagittal view (C) of the vermis (arrow) can be achieved using three-dimensional sonography. (B) Coronal view.



eFig. 9.8 Color and Spectral Doppler Interrogation of the Middle Cerebral Artery. Abnormally high peak velocity is an indicator of moderate to severe fetal anemia. Because the actual velocity is being measured, the angle of insonation should line up with the vessel.



eFig. 9.9 The three-dimensional sonography acquisition and measurement of fetal thigh volume has been used in the management of fetal growth restriction.

the detection of fetal heart anomalies appears to be limited.¹¹ The lack of technique standardization and variable operator expertise in post-processing manipulation may be barriers for widespread use of 3D ultrasound technology.¹² Additional research is needed to improve the utility of 3D ultrasound in standard obstetric practice. One major area for exploration might be the use of 3D ultrasound in the remote-read setting.

In many communities, 3D ultrasound is also available to patients in a nonmedical setting for entertainment or to provide keepsake images of the fetus. The use of ultrasound for nondiagnostic purposes has been discouraged by the US Food and Drug Administration (FDA), the ACOG, and the AIUM.^{6,12,13} Concerns raised in statements include possible adverse bioeffects of ultrasound energy, the possibility that an examination could give false reassurance to parents, and the fact that abnormalities may be detected in settings where personnel are not prepared to discuss and provide follow-up for concerning findings.

FIRST-TRIMESTER ULTRASOUND

First-trimester ultrasounds are performed before 14^{0/7} weeks of gestation. There are various indications for first-trimester ultrasound (Box 9.1).¹⁴ This type of ultrasound can be performed using a transvaginal or transabdominal approach. Often, in early gestation, transvaginal ultrasound can visualize the conceptus before the transabdominal approach due to the proximity of the probe to the pregnancy structure. The gestational age after which a pregnancy can be clearly visualized using the transabdominal approach will often vary by the pregnant person's anatomy (e.g., orientation of the uterus) and habitus, and operator experience. While many fetal anomalies (e.g., anencephaly, conjoined twins) can be diagnosed prior to 11^{0/7} weeks of gestation, most assessment for the anatomy in the first trimester happens after 11^{0/7} weeks of gestation due to an improved visualization and detection rate. Concerns regarding anatomic evaluation prior to this time frame include inability to perform diagnostic testing (i.e., chorionic villus sampling), possible introduction of unnecessary anxiety, and comparatively limited data on the diagnostic performance of ultrasound to detect anomalies prior to 11^{0/7} weeks of gestation compared to after. We may see continued efforts for earlier diagnosis of anomalies due to changes in the legal and sociocultural landscape as it pertains to access to abortion care.

BOX 9.1 Indications for First-Trimester Ultrasonography¹⁴

- To confirm the presence of an intrauterine pregnancy
- To evaluate a suspected ectopic pregnancy
- To evaluate vaginal bleeding
- To evaluate pelvic pain
- To estimate gestational age
- To diagnose or evaluate multiple gestations, including chorionicity
- To confirm cardiac activity
- As adjunct to chorionic villus sampling, embryo transfer, or localization and removal of an intrauterine device
- To assess for certain fetal anomalies, such as anencephaly, in patients at high risk
- To evaluate pelvic or adnexal masses or uterine abnormalities
- To screen for fetal aneuploidy
- To evaluate suspected hydatidiform mole

Early First-Trimester Normal Findings

A collaborative effort of multiple organizations of professionals who perform obstetric ultrasound has outlined the critical parameters of a first-trimester scan.^{6,15} The sequence of events for visualization of normal early pregnancy structures by transvaginal ultrasound is generally uniform across pregnant individuals. The gestational sac can be seen around 5 weeks of gestation, followed by a yolk sac at 5½ weeks, and lastly an embryo with fetal cardiac activity by 6 weeks (eFig. 9.10).¹⁶ **Routine use of pulsed wave or color Doppler prior to 10^{6/7} weeks is discouraged due to a theoretical risk to the embryo. Therefore M-mode ultrasound is used to document cardiac activity during this time** (see eFig. 9.6).¹⁷ Starting at 7 weeks, the embryo has grown to the point that recognizable features, such as a cephalic pole, can be seen. As shown in eFig. 9.6, a prominent midline brain vesicle can be seen at this time. The cerebral falx is visible at 9 weeks, and the appearance and disappearance of physiologic gut herniation are noted between 8 and 11 weeks (eFig. 9.11). During this physiologic process, the bowel is seen to lie within the umbilical cord and does not float freely. For this reason, the diagnosis of an abdominal wall defect should be made with caution until after 12 weeks of gestation. **Until 12 to 13 weeks, the crown-rump length (CRL) should be measured for gestational age determination.** Care should be taken to measure the full length of the fetus. The gestational age can be significantly underestimated if an oblique plane is used (eFig. 9.12).

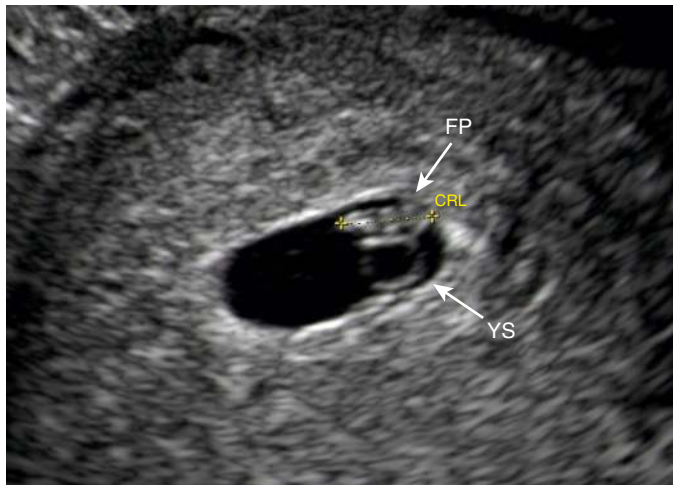
First-Trimester Anatomic Survey

The current widespread practice is to perform first-trimester ultrasounds with an assessment of basic fetal anatomy, as well as nuchal translucency to investigate fetal aneuploidy. Basic fetal anatomy ultrasounds typically include the cranium, falx, choroid plexus, midsagittal nuchal translucency view, presence of the heart with a normal axis, presence of the stomach, and presence of extremities with three segments¹⁸ (Box 9.2).

With the advent of ultrasound, attempts have been made to perform a more detailed evaluation of fetal anatomy in the first trimester. **Previous studies have reported an overall detection rate of 28% to 43% for fetal anomalies in the first trimester with a detailed assessment protocol.**^{19,20} The actual detection rate for each anomaly can vary based on the severity and by system organ. Conditions such as anencephaly, abdominal wall defects (omphalocele, gastroschisis),

BOX 9.2 Suggested Components of Basic First-Trimester Fetal Ultrasound¹⁸

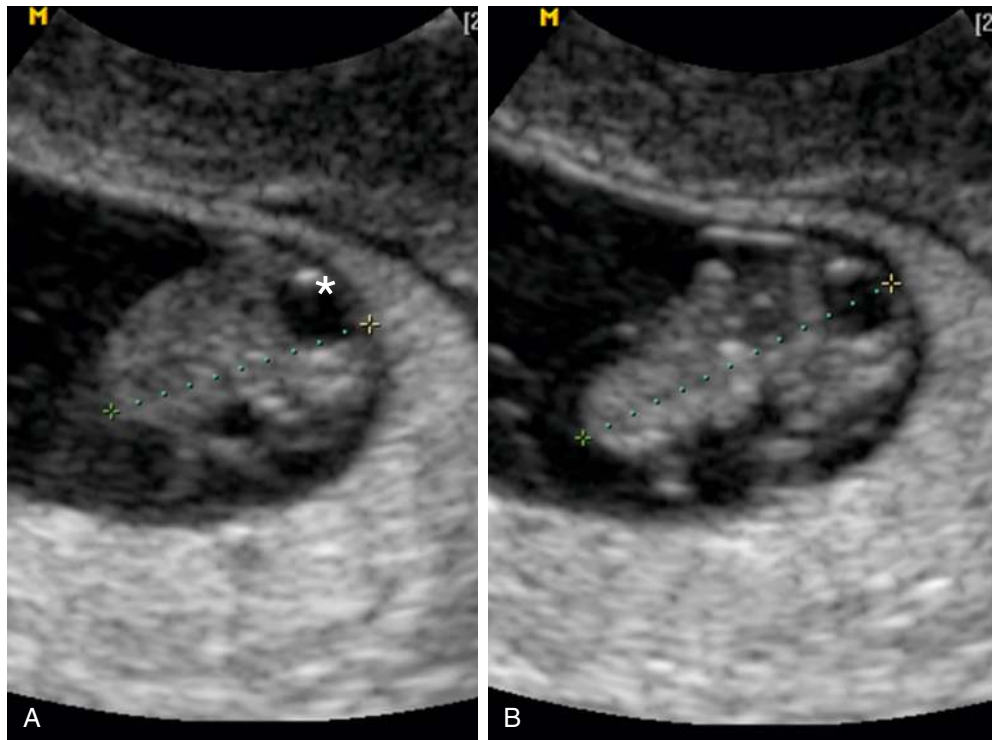
- Fetal cardiac activity (present or absent, normal or abnormal)
- Number of fetuses (if multiples, document chorionicity, amnionicity, comparison of fetal sizes, estimation of amniotic fluid normality in each sac)
- Biometry (crown-rump length [CRL] or if CRL >84 mm, biparietal diameter, head circumference, abdominal circumference, and femur length)
- Nuchal translucency
- Head and brain
 - Head shape, ossification
 - Falx present, butterfly-shape choroid plexus
- Heart
- Abdomen
- Extremities
- Placenta
- Adnexa
- Cervix



eFig. 9.10 Normal 7-Week Gestational Sac With the Yolk Sac (YS) and Adjacent Fetal Pole (FP). Calipers show the measurement to establish the gestational age. CRL, Crown-rump length.



eFig. 9.11 Nine-Week Fetus Showing the Physiologic Gut Herniation (curved arrow). Note also how the amnion (straight arrow) has not yet fused to the chorion at the uterine wall.



eFig. 9.12 (A) Image of the fetus is cut obliquely, and the crown-rump length is inappropriately short. (B) Correct measurement of crown-rump length. The difference in the calculated gestational age from these two measurements was 5 days. The normal brain vesicle (asterisk) is noted in A.

ectopia cordis, and encephalocele are almost always diagnosable in the first trimester. On the other hand, conditions such as neuronal migration disorders, duodenal atresia, and echogenic lung lesions (congenital pulmonary airway malformation or bronchopulmonary sequestration) are almost never diagnosable in the first trimester.²¹ It is prudent to remember, however, that first-trimester detailed anatomic ultrasound is not meant to replace a second-trimester assessment of fetal anatomy. There is a false-positive risk of 3% to 4% with first-trimester anatomy assessment. As such, uncertain findings need to be followed by a later assessment at 16 to 20 weeks.

Despite its promises, there are still challenges associated with the performance of first-trimester detailed anatomy scans. As with various newer technologies, there might be a learning curve for operators in performing this type of scan. Because of this, the duration of first-trimester ultrasound may be longer than when only basic first-trimester anatomy evaluation is performed. However, with increasing experience, we believe the time difference will become a nonsignificant issue.

The ISUOG updated its practice guidelines in 2023 regarding the performance of basic and detailed first-trimester ultrasound.¹⁸ **The ISUOG suggested 11 to 14 weeks of gestation as an ideal time frame for the performance of a first-trimester ultrasound.** This time frame was likely chosen as a natural extension for the already widely performed nuchal translucency ultrasound and, if desired, diagnostic testing with chorionic villus sampling (CVS) available during this time frame. Performance of the first-trimester anatomy scan can be achieved by using transabdominal or transvaginal transducers or a combination of both, depending on the particular person's uterine shape, their position, and their acoustic impedance. **The use of color, pulsed wave Doppler, and 3D ultrasound acquisition sometimes is required, and judicious use of these technologies in the first trimester is believed to be safe.**¹⁷ The ISUOG also outlined recommended standard views for first-trimester detailed anatomy (Box 9.3).

Head and Brain

The fetal head and brain in the first trimester need to be evaluated primarily in axial and sagittal views. Axial views at the level of the lateral ventricles and choroids will allow for the evaluation of the intact

cranium and its normal oval shape, normal lateral ventricle, normal falx, and choroid plexus (eFig. 9.13). Anomalies such as anencephaly, encephalocele, or alobar holoprosencephaly would be apparent on this view. Lateral ventriculomegaly is not clearly well defined in the first trimester. However, the ratio of the choroid plexus (CP) size relative to the size of the ventricle has been used to identify lateral ventriculomegaly during this period.²² Clinically, ventriculomegaly can also be diagnosed qualitatively when the choroids occupy less than half of the lateral ventricle. Additional craniofacial features, such as the fetal ear, can also be assessed in the first trimester using a slightly more oblique angle from the typical axial plane described earlier (eFig. 9.14).

Nuchal translucency can commonly be visualized and measured in the midsagittal plane. Increased nuchal translucency thickness alone is not an anomaly; it is simply a marker that could predict chromosomal aneuploidy or a congenital heart defect.^{23,24} Additionally, in this plane various assessments of facial structures can be made (Fig. 9.8). The presence and the measurement of the nasal bone can be performed in

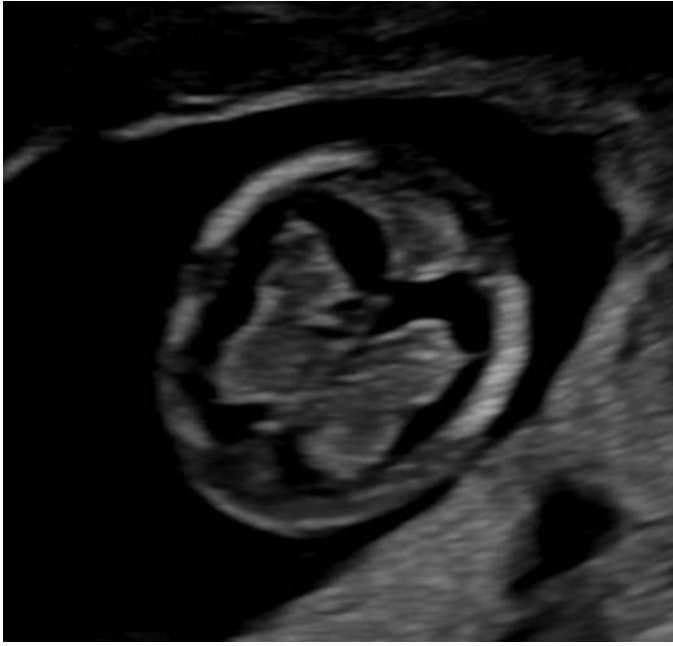


Fig. 9.8 Midsagittal View of the Fetal Cranium in the First Trimester. This view allows for visualization of multiple important structures, including nuchal translucency (NT), nasal bone (NB), maxilla (MAX), mandible (MAN), thalamus (TH), midbrain (MB), brainstem (BS), intracranial translucency (IT), and cisterna magna (CM).

BOX 9.3 Suggested Components of Detailed First-Trimester Fetal Ultrasound^{18,20}

- Fetal cardiac activity (present or absent, normal or abnormal)
- Number of fetuses (if multiples, document chorionicity, amnionicity, comparison of fetal sizes, estimation of amniotic fluid normality in each sac)
- Biometry (crown-rump length [CRL] or if CRL >84 mm, biparietal diameter, head circumference, abdominal circumference, and femur length)
- Nasal bone
- Ductus venosus^a
- Tricuspid valve regurgitation^a
- Uterine artery pulsatility index^a
- Head and brain
 - Head shape, ossification
 - Falx present
 - Butterfly-shape choroid plexus
 - Intracranial translucency/brainstem/cisterna magna
 - Cerebral peduncles with aqueduct of Sylvius
- Face and neck
 - Nuchal translucency
 - Retronasal triangle
 - Maxilla/mandible
 - Orbits
- Heart
 - Heart intrathoracic with regular rhythm
 - Cardiac size and axis
 - Four-chamber view
 - Left ventricular outflow tract
 - Right ventricular outflow tract
 - Three-vessel trachea view
- Abdomen
 - Stomach filled
 - Bladder filled (length <7 mm)
 - Abdominal cord insertion with intact abdominal wall
 - Umbilical arteries
 - Kidneys (coronal)
- Spine (sagittal view)
- Extremities
 - Upper limbs with three segments
 - Lower limbs with three segments
- Placenta
- Cervix

^aAs indicated.



eFig. 9.13 Axial view of the fetal cranium in the first trimester demonstrating intact cranium, symmetrical choroid plexus, and intact falx cerebri.



eFig. 9.14 Oblique view of the fetal face demonstrating bilateral ears in the first trimester (*white arrows*).

this view to assess for the risk of aneuploidy. The maxilla will appear as a rectangular hyperechoic shape, and a gap in this rectangle can be a sign of cleft palate. A quantitative assessment of the mandible can be made to evaluate for micrognathia or retrognathia. Assessment of the maxilla and mandible can be enhanced with an oblique coronal view of the face with visualization of the retronasal triangle and mandibular gap. Neurosonography in the first trimester is particularly challenging due to rapid developmental changes during this period, as well as sonographic resolution limitation.²⁵ Nonetheless, various structures of the fetal central nervous system can be assessed in the first trimester using the standard midsagittal view (see Fig. 9.8). The deviation of these structures from normal can be a first-trimester sonographic sign for an open neural tube defect with a sensitivity of nearly 80% and specificity of over 98%.²⁶

Thorax and Heart

First-trimester assessment of the fetal heart began three decades ago. Since this time, the sensitivity of ultrasonographic technology for the detection of congenital heart defects continues to increase.²⁷ A meta-analysis of studies on first-trimester detection of congenital heart defects showed a pool sensitivity of 56%. Studies within this meta-analysis represented nearly 80% of all prenatally diagnosed major congenital heart defects.²⁸ Various screenings of cardiac views in the second trimester can also be obtained in the first trimester. Screening views in the first trimester typically include an axial four-chamber heart with and without color, and a three-vessel trachea view with and without color. In the four-chamber heart view, assessment of the cardiac axis (between 30 and 60 degrees), relative ventricular size, ribs, lungs, and situs can be made (Fig. 9.9). The three-vessel trachea view primarily allows for the assessment of the alignment and size of the outflow tracts. These views are often easier to obtain using color Doppler, which can enhance the reliability of these assessments (Fig. 9.10). Major congenital heart defects such as atrioventricular canal defect and hypoplastic left or right heart syndrome can be diagnosed in the first trimester (eFig. 9.15). Additional views of the left and right ventricular outflow tracts and pulmonary veins can also be

reliably obtained in the first trimester, especially when microvascular Doppler technology is utilized (eFig. 9.16).^{29,30} In the sagittal plane, ductal and aortic arches and the bicaval view can be reliably obtained with color Doppler (eFig. 9.17). Additionally, the coronal view can sometimes be used to ascertain an intact diaphragm. Using this view, some cases of congenital diaphragmatic hernia can be diagnosed in the first trimester.

In some cases, acquisition of all cardiac views may not be possible during early gestation. Quarello and colleagues have previously suggested that at the very least, four-chamber heart and three-vessel trachea views are sensitive markers in screening for congenital heart defects.³¹ Nonetheless, some congenital heart defects continue to evolve as pregnancy progresses. As such, some lesions may not be visually apparent until the second trimester or in some cases, the third trimester.

Abdomen

During the window for first-trimester detailed anatomy assessment, the stomach and bladder should be clearly visualized. The stomach should be located in the upper left abdomen, and levocardia and correct situs should be confirmed. As opposed to the second trimester, visualization of kidneys in the first trimester is best achieved in a coronal plane. In this view, the presence and the cortical echotexture of their parenchyma can be assessed (eFig. 9.18). In some cases, the presence of bilateral renal arteries can be demonstrated using color Doppler (eFig. 9.19). The bladder can be reliably viewed in sagittal, axial, or coronal planes. The size and texture of the bladder wall should be assessed; normal bladder diameters are typically less than 7 mm. In the axial view of the bladder, the number of umbilical arteries can also be documented with color Doppler (eFig. 9.20). The integrity of the abdominal wall typically is documented with a view of a normal abdominal cord insertion (eFig. 9.21). Physiologic midgut herniation can be seen up to 11 weeks, and repeat evaluation is recommended if omphalocele is suspected before this gestational age (eFig. 9.22). Finally, while fetal sex can be predicted with the orientation of the genital tubercle in the first trimester, the routine performance of fetal sex prediction is not recommended as its clinical utility is debatable.

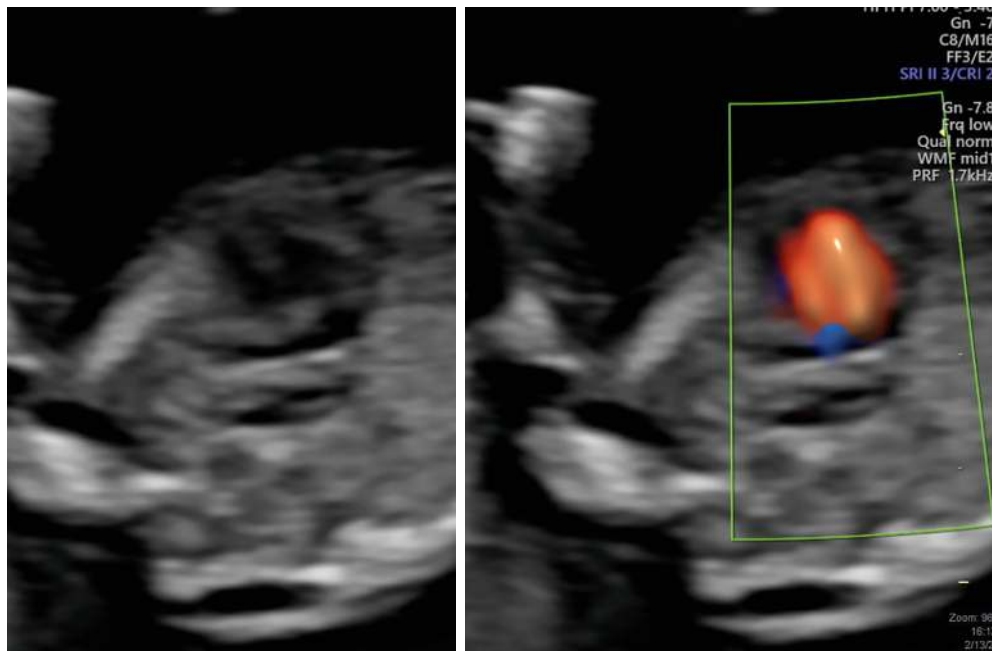
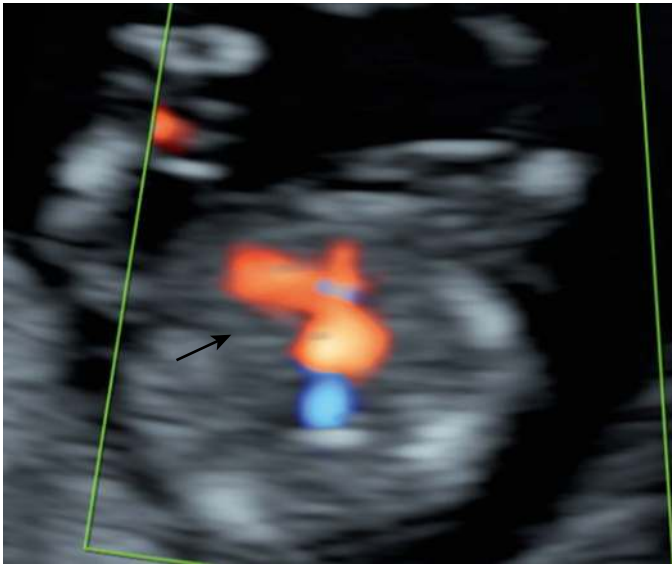


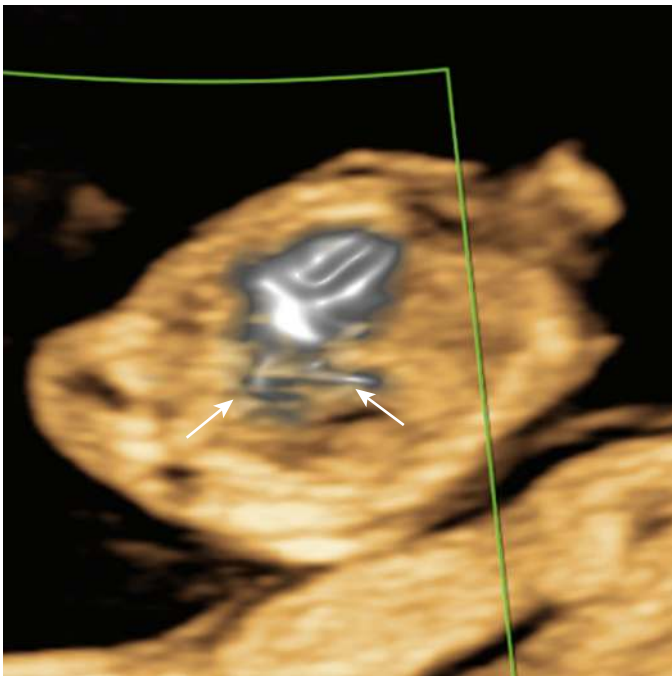
Fig. 9.9 Axial view of fetal thorax demonstrating four-chamber heart view with and without color Doppler.



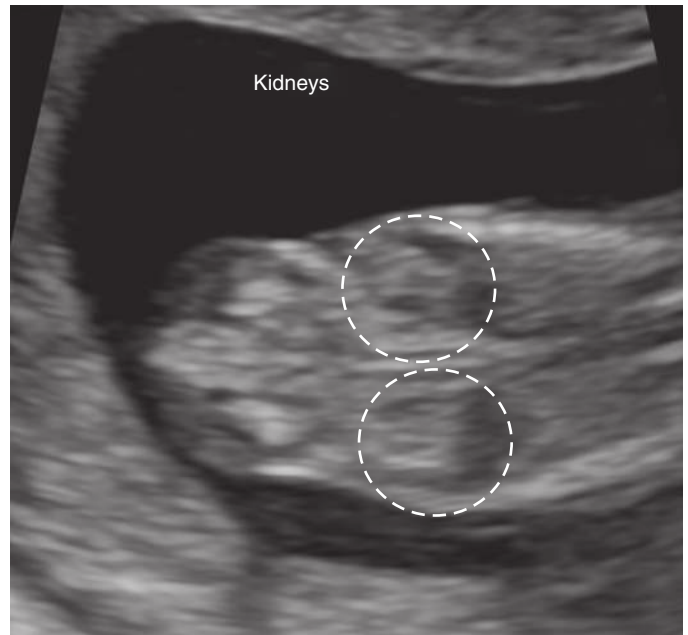
eFig. 9.15 Axial view of the fetal thorax showing four-chamber view in a case of hypoplastic left heart syndrome in the first trimester. The left ventricle was hypoplastic as demonstrated by the lack of color flow in this area (*black arrow*).



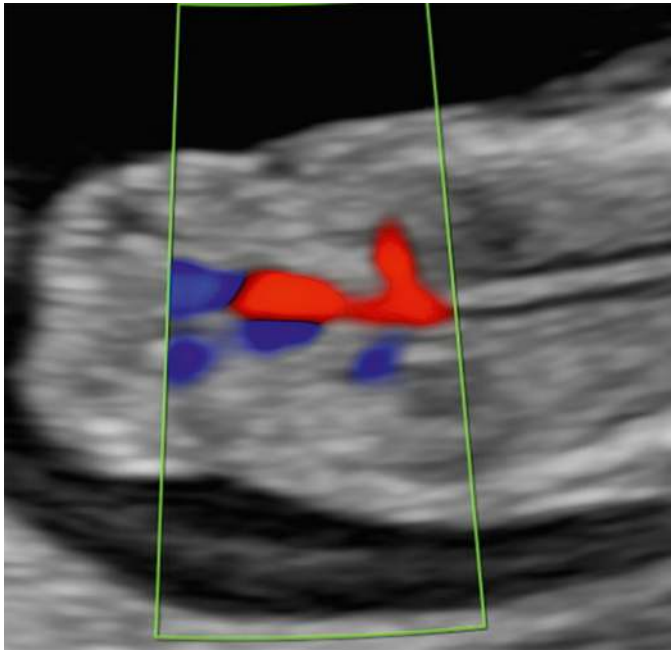
eFig. 9.17 Sagittal view of the aortic arch in the first trimester with color Doppler in the case of hypoplastic left heart syndrome. Notice the flow reversal in the ascending aorta toward the hypoplastic left ventricle.



eFig. 9.16 Four-Chamber View in the First Trimester. With the use of microvascular Doppler, pulmonary veins can be seen easily (*arrows*).



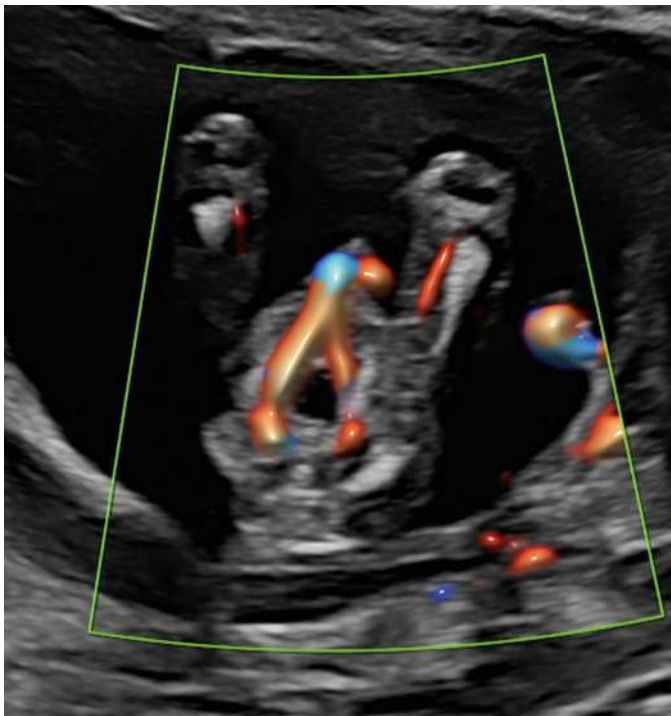
eFig. 9.18 Coronal View of the Fetal Abdomen in the First Trimester. Fetal kidneys can be seen (*dashed circles*).



eFig. 9.19 Color Doppler of the fetal abdomen demonstrating the presence of bilateral renal arteries in the first trimester.



eFig. 9.21 Axial view of the fetal abdomen in the first trimester at the level of the umbilical cord insertion. Notice the intact abdomen to the left and the right of the cord insertion (*arrow*).



eFig. 9.20 Axial View of the Fetal Bladder in the First Trimester. With concurrent use of color Doppler, the presence of two umbilical arteries can be documented.



eFig. 9.22 Three-dimensional surface rendering view of 10-weeks' gestation fetus with physiologic midgut herniation (*dotted line*).

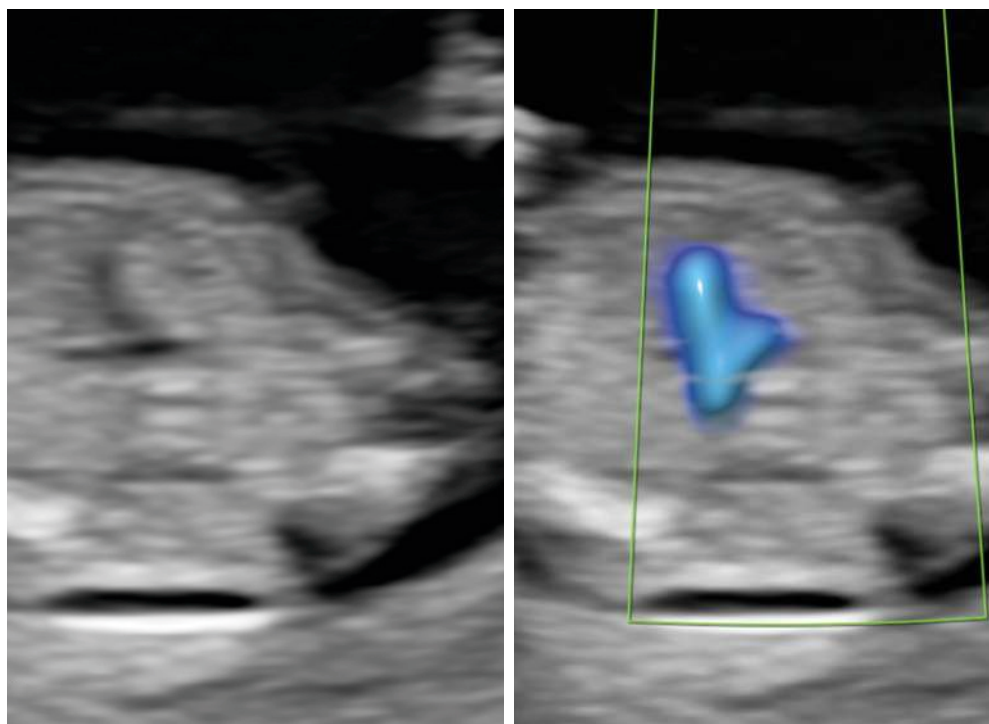


Fig. 9.10 Axial view of fetal upper thorax demonstrating three-vessel trachea view (*left*). Visualization of this view can be assisted by using color Doppler (*right*).

Spine

The curvature and alignment of the vertebral bodies can be seen on ultrasound starting around 12 weeks after their ossification. Given the relatively smaller size of the vertebral bodies in the first trimester, a single sagittal view can often be used to evaluate the entire length of the spine. Visualization of the spine will allow for the diagnosis of open neural tube defects. Similar to the second-trimester assessment, visualization of normal intracranial neuroanatomy can aid help in assessing for normal spinal anatomy.³²⁻³⁴ While various intracranial markers are used for the detection of spina bifida in the first trimester, the underlying concept is similar, involving examination of the relative posterior-caudal displacement of intracranial structures.

Limbs

Extremities can be documented with three segments for each upper and lower limb along with the alignment of hands and feet in the first trimester (eFigs. 9.23 and 9.24). Depending on the obtained resolution, though not always possible, the number of digits can also be visualized during this time frame. The bone echogenicity and appearance also need to be noted, as some forms of severe skeletal dysplasia can be present at this early stage of gestation.

SECOND- AND THIRD-TRIMESTER ULTRASOUND

Types of Examinations

Major professional organizations that guide the use of ultrasound in obstetrics have defined a set of criteria for standard obstetric ultrasound examinations performed in the second and third trimesters.^{15,35,36} These guidelines recognize that not all ultrasound examinations have the same purpose. A **standard** second- or third-trimester examination (Current Procedural Terminology [CPT] code

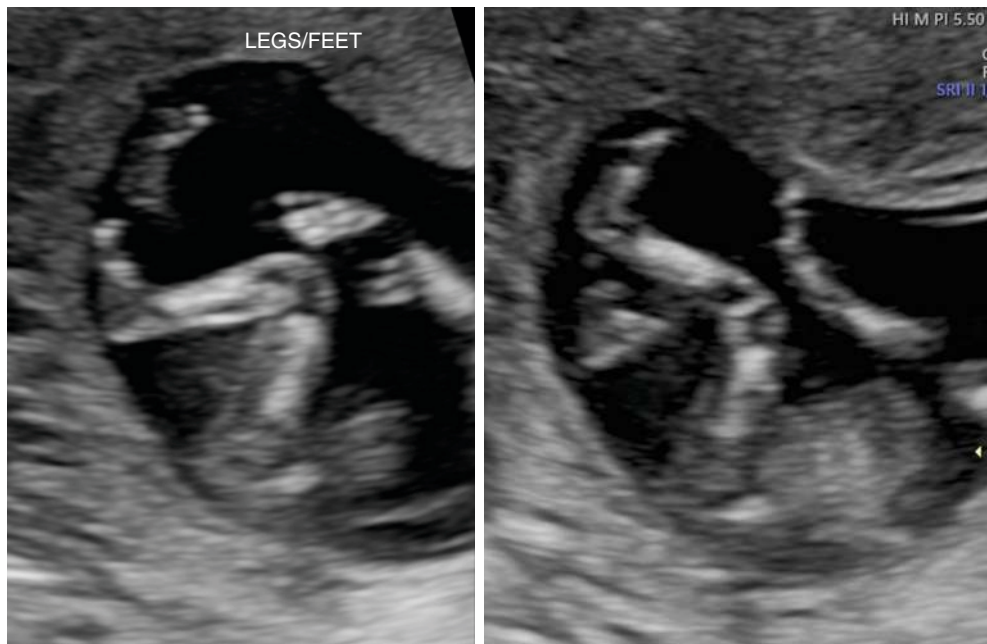
76805), as defined in Box 9.4, can be performed by any appropriately qualified sonographer. It is recognized that certain scans, termed *specialized* examinations (CPT code 76811), are more complex than the complete standard examinations performed during the course of routine pregnancy care. These examinations are commonly referred to as detailed anatomic surveys. **This designation and its billing code are intended to be used by referral practices with special expertise in the identification of and counseling on fetal anomalies.**^{6,35} A detailed survey typically contains the elements defined in Box 9.5. These elements can be expanded as necessary, depending on the clinical indication. For example, additional views of fetal neuroanatomy, including corpus callosum or parasagittal view of the lateral ventricles, may be necessary for patients who are referred for suspected fetal intracranial abnormalities. Aside from the detailed survey, other specialized examinations may include fetal Doppler ultrasonography and fetal echocardiography. Follow-up examinations are needed for many obstetric conditions; these are termed **repeat** examinations (CPT code 76816).

Another examination category is the *limited* examination (CPT code 76815). Limited ultrasound examinations, also performed by an individual with appropriate training, are used to obtain a specific piece of information about the pregnancy. Examples include the determination of fetal cardiac activity and fetal lie, assessment of amniotic fluid volume, or the measurement of cervical length. Unfortunately, it is all too common for practitioners to perform limited examinations in a manner inconsistent with good medical practice. For example, in some clinics practitioners perform an ultrasound at the first prenatal visit to document viability but do not measure the fetus or record the results of the examination. Such a practice can create problems later in pregnancy when the gestational age is in doubt.

All the aspects of the standard obstetric examination listed in Box 9.4 are important for clinical management and need to be kept in mind during every examination. A description of the importance of



eFig. 9.23 First-Trimester View of Fetal Upper Extremities. The presence of three segments of the upper extremities and their alignment were clearly demonstrated.



eFig. 9.24 First-Trimester View of Fetal Lower Extremities. The presence of three segments of the lower extremities and their alignment were clearly demonstrated.

BOX 9.4 Suggested Components of the Standard Obstetric Ultrasound Performed in the Second and Third Trimesters^{6,35}

- Standard biometry
 - Fetal cardiac activity (present or absent, normal or abnormal)
 - Number of fetuses (if multiples, document chorionicity, amnionicity, comparison of fetal sizes, estimation of amniotic fluid normality in each sac, and fetal genitalia)
 - Presentation
 - Qualitative or semiquantitative estimate of amniotic fluid volume
 - Placental location, especially its relationship to the internal os; placental cord insertion site if technically possible
 - Evaluation of the uterus, adnexal structures, and cervix when clinically appropriate and technically feasible
- Anatomic Survey to Include:**
- Head and neck
- Midline falx
 - Cavum septum pellucidum
 - Lateral cerebral ventricles
 - Choroid plexus
- Cerebellum
 - Cisterna magna
 - Fetal lip
 - Nuchal skin fold may be helpful for aneuploidy risk
- Chest
- Four-chamber view of the heart
- Left and right ventricular outflow tracts
- Abdomen
- Stomach (presence, size, and situs)
 - Kidneys
 - Urinary bladder
 - Umbilical cord insertion into the abdomen
 - Number of umbilical cord vessels
- Spine
- Extremities (presence or absence of legs and arms)
- Genitalia

BOX 9.5 Suggested Components of the Detailed Obstetric Ultrasound Performed in the Second and Third Trimesters^{6,35}

- Standard biometry
 - Fetal cardiac activity (present or absent, normal or abnormal)
 - Number of fetuses (if multiples, document chorionicity, amnionicity, comparison of fetal sizes, estimation of amniotic fluid normality in each sac, and fetal genitalia)
 - Presentation
 - Qualitative or semiquantitative estimate of amniotic fluid volume
 - Placental location, especially its relationship to the internal os; placental cord insertion site if technically possible
 - Evaluation of the uterus, adnexal structures, and cervix when clinically appropriate and technically feasible
- Anatomic Survey to Include:**
- Head and neck
- Cranium
 - Midline falx
 - Cavum septum pellucidum
 - Lateral cerebral ventricles
 - Choroid plexus
 - Cerebellum, vermis, and cisterna magna
 - Fetal lip, palate, maxilla, mandible, and tongue
 - Profile with nasal bone
 - Orbits
 - Nuchal skin
- Heart
- Situs
 - Four-chamber view
 - Left ventricular outflow tract
 - Right ventricular outflow tract
 - Three-vessel view
 - Three-vessel trachea view
 - Ductal arch
 - Aortic arch
 - Superior and inferior venae cavae
 - Interventricular septum
- Thorax
- Lungs
 - Diaphragms
- Abdomen
- Stomach (presence, size, and situs)
 - Gallbladder
 - Kidneys and renal arteries
 - Urinary bladder
 - Umbilical cord insertion into the abdomen
 - Number of umbilical cord vessels
- Spine (at least in sagittal and coronal views)
- Extremities (including digits of hands and feet)
- Genitalia

the components of the standard ultrasound examination is provided in a later section.

Qualifications for Performing and Interpreting Diagnostic Ultrasound Examinations

Most ultrasound exams in the United States are performed by professionals who are credentialed by the American Registry of Diagnostic Medical Sonography (ARDMS). Individuals with these credentials

have had extensive education and testing to ensure their competency. Following the ultrasound exam, a physician will generate a report based on the images and information obtained by the sonographer. Often the physician (sonologist) may personally perform or repeat parts of the exam. In a practice bulletin, the ACOG set forth criteria by which licensed medical practitioners are qualified to perform or interpret ultrasound examinations.⁶ These individuals should be versed in the indications and expected content of a complete exam and

should have knowledge regarding the limitations of diagnostic ultrasound. They should also understand the anatomy and pathophysiology of structures being evaluated and should have specific training in obstetric sonography. Additionally, these individuals are responsible for the quality, accuracy, documentation, and safety of exams performed under their supervision.

COMPONENTS OF THE EXAMINATION

Cardiac Activity

The presence or absence of cardiac activity should be documented and is usually the starting point of an exam. As noted previously, after about 6 weeks' gestation, the diagnosis of fetal life is rarely difficult. Even though fetal death may be obvious with B-mode imaging, confirming the absence of a heartbeat with color or pulsed wave Doppler is recommended. Throughout pregnancy, an abnormally fast, slow, or irregular heartbeat can be detected by visual inspection using gray-scale ultrasonography. The abnormal rate can be quantified and documented with M-mode ultrasound (see Fig. 9.3) or pulsed wave Doppler ultrasound (Fig. 9.11).

Number of Fetuses

When a multiple pregnancy is diagnosed, the number of amnions and chorions should always be determined (see Chapter 39). Determination of chorionicity is most easily accomplished in early pregnancy (Fig. 9.12). The presence of unlike sex, separate placentas (eFig. 9.25), or a thick membrane dividing the sacs with a twin peak or *lambda sign* (Fig. 9.13) all indicate the presence of two chorions. With monozygotic twins, there is a thin intervening membrane, which attaches to the placenta without a peak (Fig. 9.14). The level of fetal risk is much higher when fetuses share a chorion, and the risk is extremely high if there is a single amnion (Fig. 9.15, eFig. 9.26).^{3,23} Monozygotic pregnancies require an early referral for a specialized ultrasound. Twin–twin transfusion syndrome (TTTS) occurs in 15%

of monozygotic pregnancies and is associated with a high degree of morbidity and mortality. **Because TTTS may have an abrupt onset and is a treatable condition, most authorities recommend ultrasound every 2 weeks, starting at 16 weeks.**³⁷ In all twin pregnancies,

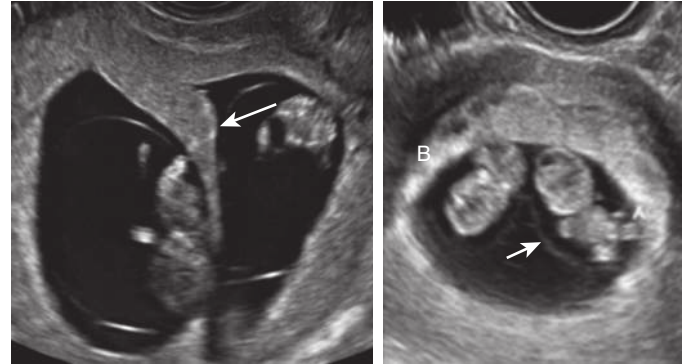


Fig. 9.12 First-Trimester Twin Pregnancies. The one on the left is dichorionic diamniotic and the one on the right is monozygotic diamniotic. In the dichorionic pregnancy, note the very prominent division between the sacs, with thick chorionic material and membrane (left arrow). In contrast, there is a wispy dividing membrane (right arrow) with the monozygotic diamniotic placentation.

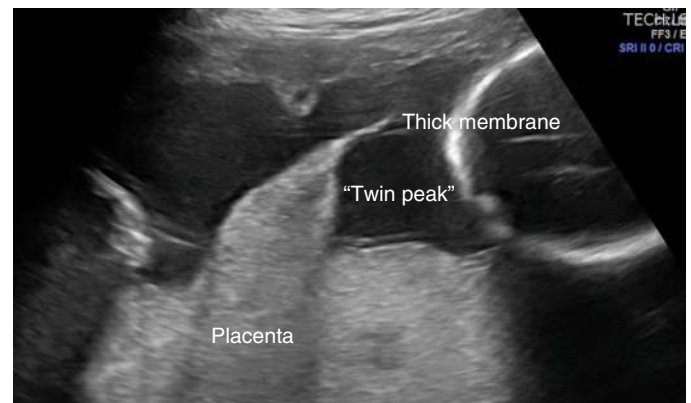


Fig. 9.13 Dichorionic Diamniotic Twin Pregnancy With a Single Posterior Placenta and a Prominent Twin Peak Sign (Also Known as Lambda Sign). There is a thick intervening membrane.

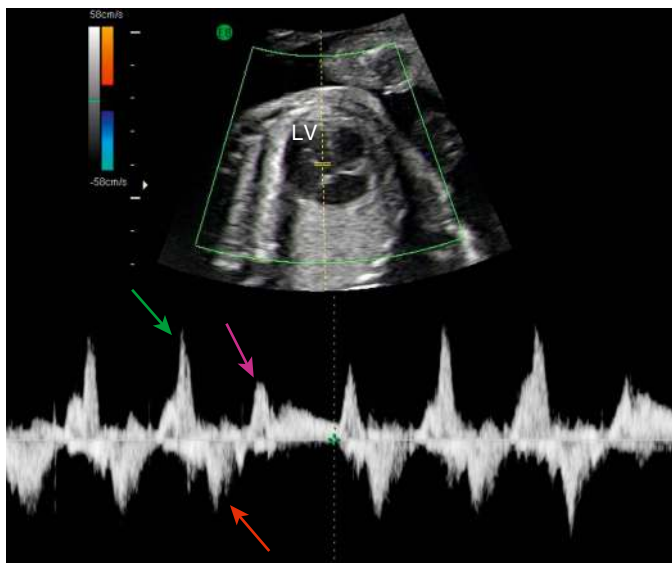
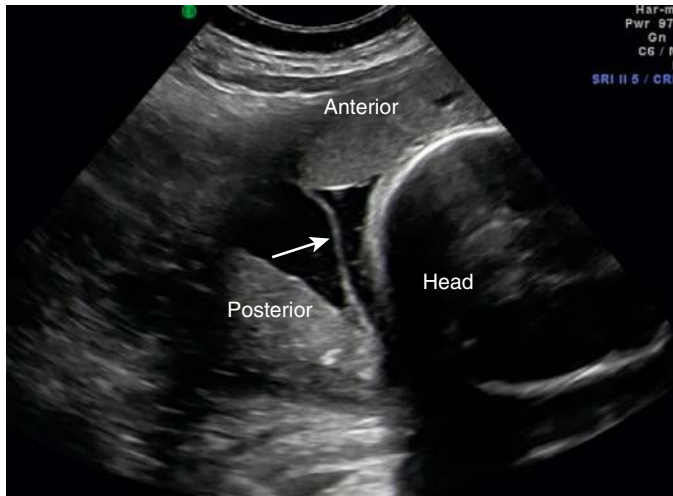


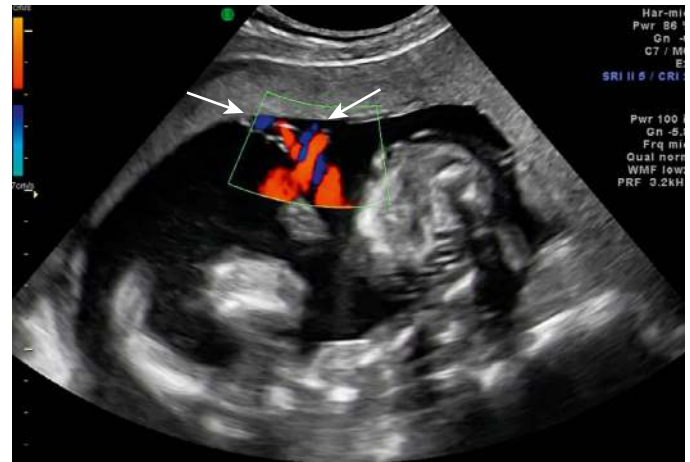
Fig. 9.11 Pulsed Wave Doppler Showing a Premature Atrial Contraction. The Doppler gate is placed in the left ventricle (LV). Flow into (green arrow) and out of (red arrow) the ventricle are displayed. There is a premature atrial beat (magenta arrow) that results in flow into the ventricle, but there is no corresponding ventricular beat, since the ventricle is in a refractory state.



Fig. 9.14 Monozygotic Diamniotic Twin Pregnancy With a Single Anterior Placenta and a Thin Intervening Membrane (small arrow). Note the absence of a twin peak sign where the membrane inserts onto the placenta (large arrow).



eFig. 9.25 Twin Pregnancy With Two Placentas, One Anterior and One Posterior. With separate placentas and a thick intervening membrane (arrow), this must be a dichorionic diamniotic pregnancy. One of the fetal heads is shown.



eFig. 9.26 Monochorionic Monoamniotic Twin Pregnancy, With Umbilical Cords Intertwined. Arrows indicate the placental cord insertions. Cord entanglement occurs with most monoamniotic twins.

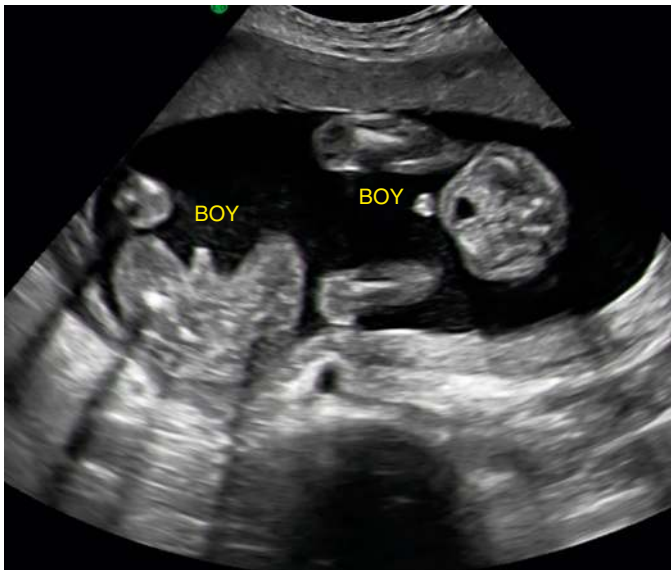


Fig. 9.15 Monochorionic Monoamniotic Twin Pregnancy. Note the absence of a membrane between the fetuses. One must be very careful in making this diagnosis, since with monochorionic diamniotic twins the membrane is very thin and may be difficult to see.

periodic ultrasound examinations should be performed to assess for fetal growth. Twins are at a significantly increased risk for growth abnormalities, and, obviously, it is not possible to assess the growth of the twins individually by abdominal palpation.

Presentation

The assessment of presentation is not merely a matter of determining whether the fetus is head down or breech. A more precise ultrasound analysis of presentation is important in certain circumstances. A **back-down transverse lie makes cesarean delivery more difficult and sometimes requires a classical incision, and a back-up transverse lie increases the risk of cord prolapse with premature rupture of membranes.** In the intrapartum setting, ultrasound can be used to reliably assess fetal head position, station, and attitude using the combination of transabdominal and transperineal ultrasound.³⁸ The knowledge of these elements may be helpful in the management of prolongation or arrest of the first and the second stages of labor.

Amniotic Fluid Volume

Every ultrasound examination should include an assessment of the amniotic fluid volume (see [Chapter 28](#)). It is acceptable for an experienced examiner to make this determination subjectively.^{6,35} However, to aid in communication and to provide criteria for management protocols, semiquantitative methods are preferred. The amniotic fluid index (AFI) is the sum of the measurements of the deepest vertical pocket (DVP) of fluid in each of the uterine quadrants ([Fig. 9.16](#)). The limits of the quadrants are the pregnant patient's midline and a

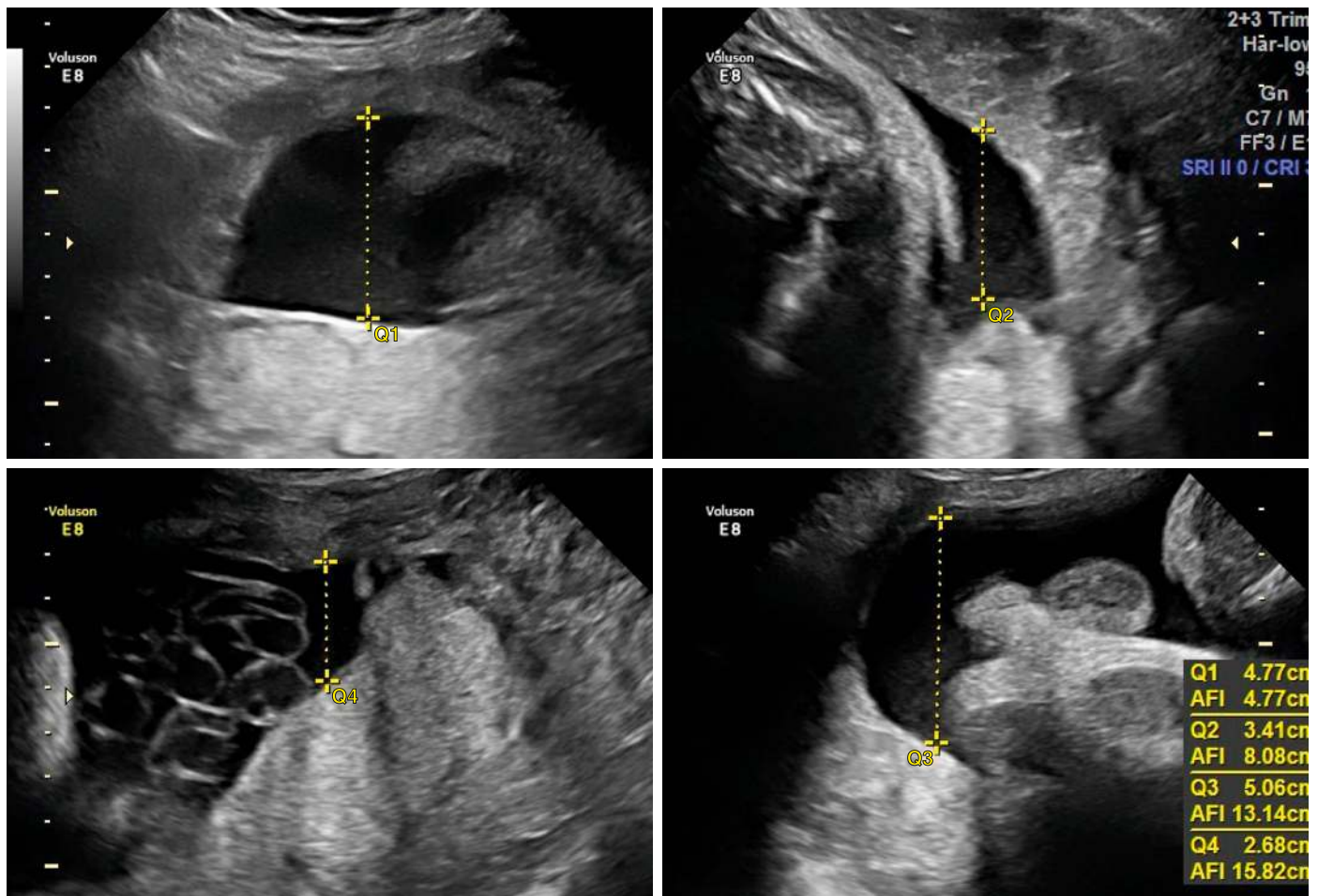


Fig. 9.16 Sonographic Images Showing the Deepest Vertical Pocket in Each of the Four Quadrants of the Uterus. The sum of the measurements of these pockets is the amniotic fluid index.

horizontal line through the patient's umbilicus. Each pocket should measure at least 1 cm in width. The line between the calipers should not cross through loops of cord or fetal parts. Polyhydramnios and oligohydramnios can be defined by either the AFI or the measurement of the single DVP. Polyhydramnios is usually defined as an AFI greater than 24 cm or DVP greater than 8 cm, and oligohydramnios as an AFI less than 5 cm or DVP less than 2 cm.^{6,39}

Placenta and Umbilical Cord

One of the principal advantages of routine ultrasound is that serious problems of placentation—such as placenta previa, placenta accreta, and vasa previa—can be diagnosed in a timely manner (see Chapter 20). At the time of the routine screening ultrasound (beyond 18 weeks), it should be determined whether the placenta covers the internal cervical os (Fig. 9.17). If the placenta and the cervix are not seen clearly (eFig. 9.27), or if it appears that the edge of the placenta is close to the cervix, transvaginal ultrasound should be used liberally to clarify this relationship (Figs. 9.18–9.21).^{6,40}

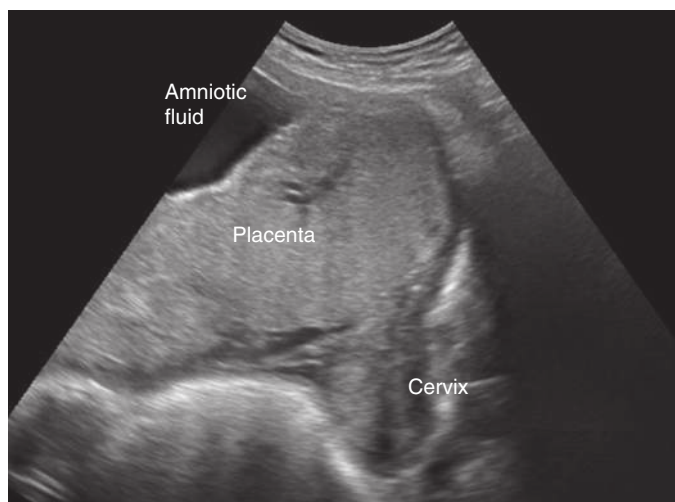


Fig. 9.17 Transabdominal Midline Sagittal View of the Lower Part of the Uterus. The placenta is centered over the cervix.

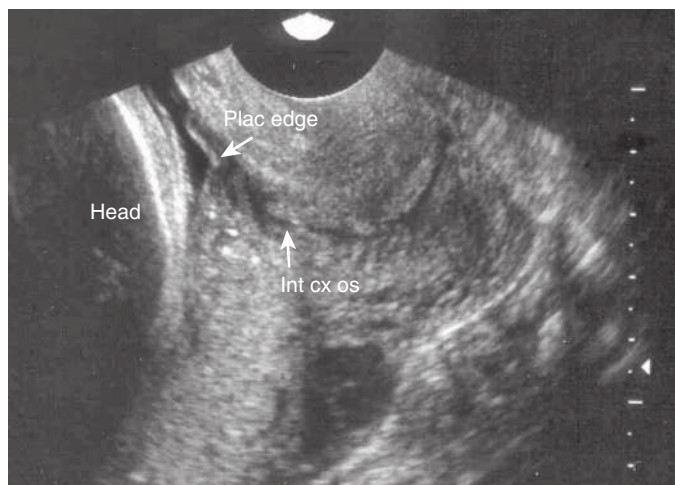


Fig. 9.18 Sagittal Transvaginal View in the Pregnancy That Clarifies the Suboptimal View Shown in eFig. 9.27. Marginal placenta is clearly visible, and the edge of the placenta (*Plac edge*) extends 1.5 cm over the internal cervical os (*Int cx os*).

Placenta Morphology

In addition to determining the placenta's location, its appearance should be assessed. Venous lakes, which can be quite large (Video 9.1),

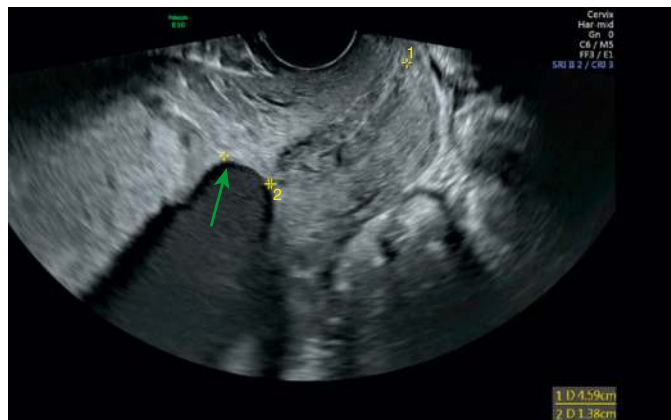


Fig. 9.19 Transvaginal Midline Sagittal View of an Anterior Low-Lying Placenta. The edge of the placenta (*green arrow*) ends 1.4 cm from the cervix. This image also shows a normal cervical length.

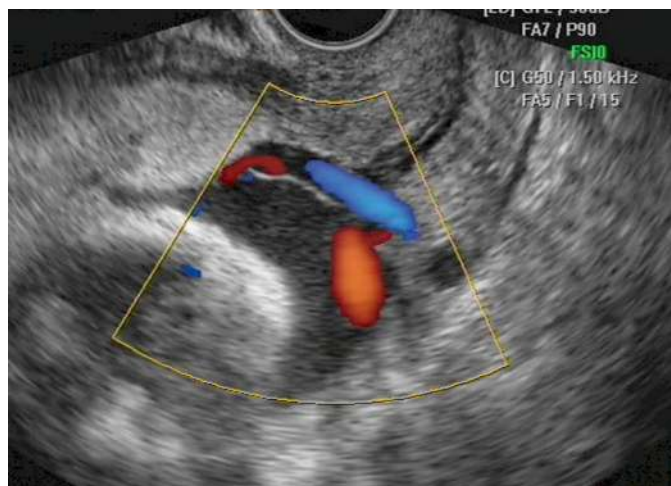


Fig. 9.20 Sagittal Transvaginal Image Showing Vasa Previa. An umbilical vessel passes over the internal os of the cervix. In this case, a velamentous cord insertion was present, and this vessel passed over the cervix, running from the cord insertion to the placenta.

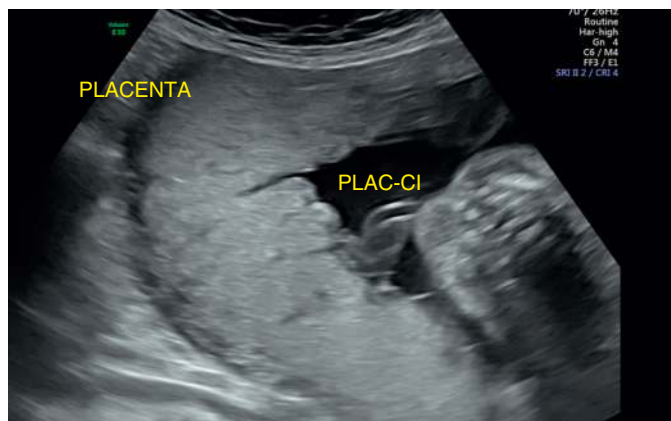
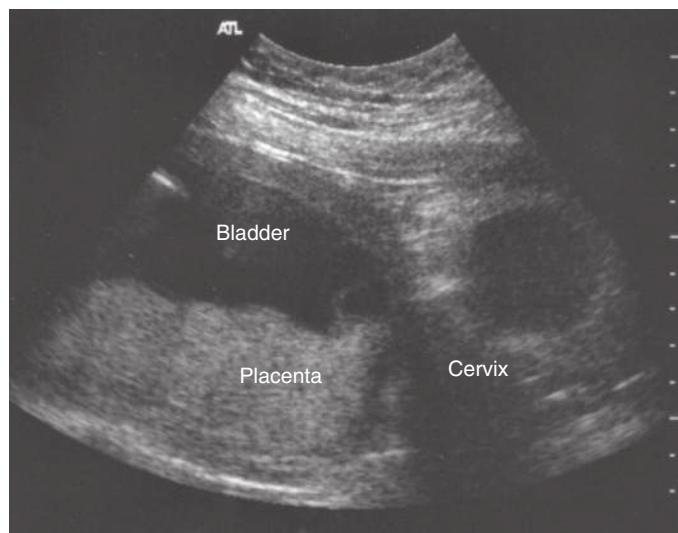


Fig. 9.21 Transabdominal Midline Sagittal View Showing the Placenta Cord Insertion (*PLAC-CI*).



eFig. 9.27 Sagittal Transabdominal View of the Lower Uterus and Cervix. The relationship between the edge of the placenta and cervix is unclear, and placenta previa cannot be accurately diagnosed. Transvaginal ultrasound resolved this uncertainty (see Fig. 9.18).

often have no clinical significance. Many changes observed in the placenta are related to calcification, fibrosis, and infarction. The general trend is for these changes to become more apparent as pregnancy progresses, but their clinical significance is unclear. Other placental morphologic differences, such as a “globular” placenta (eFig. 9.28) or a circumvallate placenta, have been associated with an increased rate of FGR, stillbirth, and other complications.^{41,42} For these patients, third-trimester growth ultrasound can be considered.

Umbilical Cord

In all pregnancies, the sonographer should confirm that there are two arteries and a vein in the umbilical cord.^{6,39} In late pregnancy, this can be ascertained by looking at a transverse cut of the cord in a free loop. In the second trimester, two umbilical arteries are most easily confirmed by identifying the vessels as they course around the fetal bladder (Fig. 9.22). A single umbilical artery is present in about 0.5% of all newborns. Since there is a 20% incidence of associated malformations, including aneuploidy, this finding should prompt a detailed fetal survey. **Fetuses with a single umbilical artery have a 20% chance of growth restriction.** Additionally, the rate of polyhydramnios, abruption, placenta previa, structural placental abnormalities, cesarean delivery, low Apgar scores, and fetal death are all increased.⁴³ For this reason, follow-up growth ultrasounds and testing for fetal well-being should be considered.

Uterus

With any obstetric ultrasound, including those performed in the first trimester, the adnexal and uterine morphology should be evaluated. Many people enter pregnancy without being aware that they have fibroids or a Müllerian malformation. Fibroids are usually readily apparent with transvaginal or transabdominal ultrasound. A common pitfall, however, is to confuse uterine contractions with fibroids (Fig. 9.23). Contractions have a more lenticular shape and blend with the surrounding myometrium, whereas fibroids are usually

more spherical with distinct borders and a whorled internal echo texture.

Cervix

It has been clearly documented that a short cervix, as measured with transvaginal ultrasound (Fig. 9.24), is associated with an increased risk of preterm birth in both high- and low-risk patients (see Chapters 35 and 36). Intervention with a cervical cerclage has been shown to improve outcomes in patients with a prior spontaneous preterm birth who have a short cervix in a subsequent pregnancy.⁴⁴ Treatment of low-risk patients who have a short cervix with vaginal progesterone has also been shown to reduce the rate of preterm birth.^{45,46} However, the value of universal screening with transvaginal ultrasound is controversial, with several studies reporting modest reduction in preterm birth.^{47,48} However, given some uncertainties in its benefit, the Society for Maternal-Fetal Medicine (SMFM) guidelines state that cervical length screening in singleton gestations without prior preterm birth is reasonable, but is not required.⁴⁹ **The ACOG currently does not recommend this practice, but does recommend that transvaginal or transperineal ultrasound be considered if the cervix appears shortened on abdominal ultrasound.**⁶ The lower limit of normal for cervical length in the mid part of the second trimester is 25 mm. Correct technique of cervical length measurement—which requires training—is important for proper clinical decision making.

Adnexae

Normally sized ovaries may be difficult or impossible to see in the second or third trimester because of shielding by the bowel. However, when a significant tumor is present, the ovaries will often be visible. Masses that consist of simple cysts usually represent benign processes and do not require removal during pregnancy. Concerning features, such as large size, multiple cystic cavities, thick septae, internal papillae, or solid areas, warrant careful evaluation and occasionally may require operative removal. The most common neoplasms in

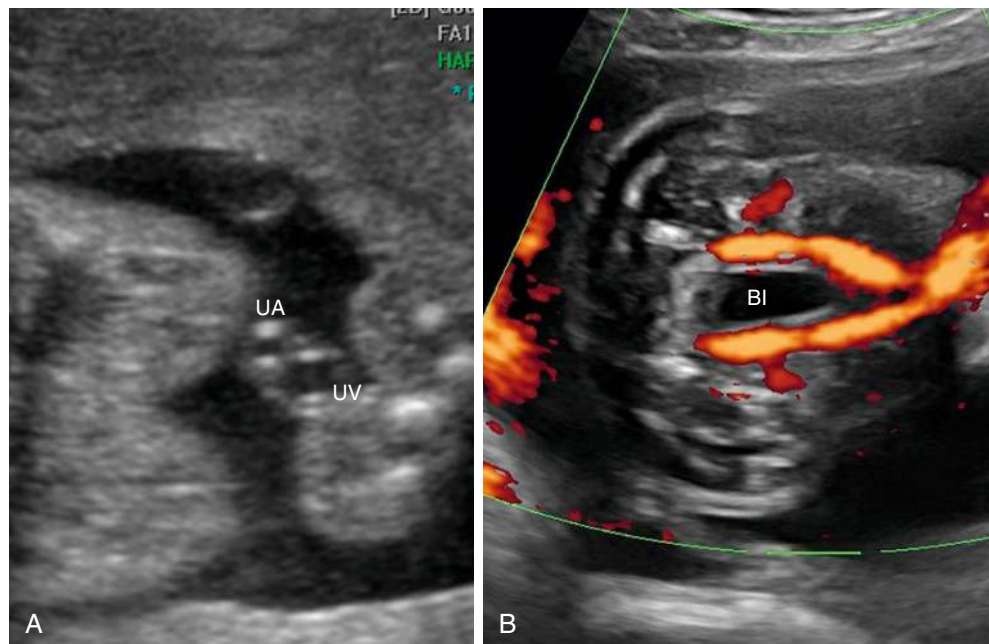


Fig. 9.22 (A) Pregnancy with a single umbilical artery (UA). The artery and umbilical vein (UV) are shown in a transverse section of a free loop of cord. (B) Transverse view of the fetal pelvis, using color Doppler, clearly shows the normal paired umbilical arteries coursing around the bladder (BI). It is often easier to document the umbilical arteries in early pregnancy with this color Doppler method.



eFig. 9.28 Globular Placenta. This type of placenta is relatively thick compared to the length of its base.

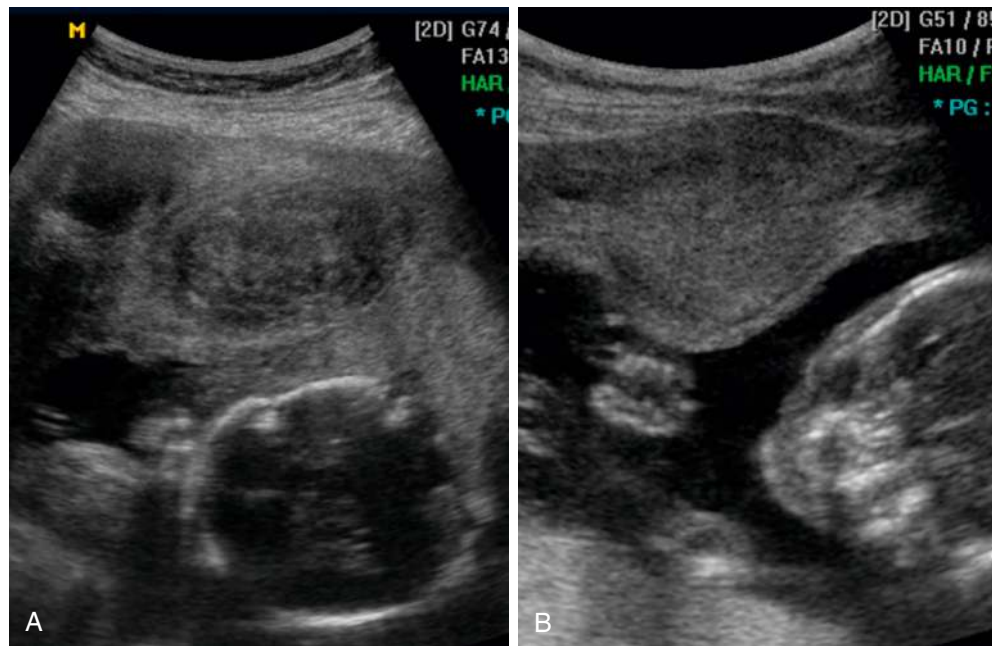


Fig. 9.23 (A) Two round and well-circumscribed fibroids. (B) Uterine contraction is lenticular in shape and does not have a clear border. Contractions such as this are very common in the second trimester.



Fig. 9.24 Midline Sagittal Transvaginal Ultrasound Showing Funneling of the Cervix (*Arrow*). The closed portion of the cervix measures 16 mm (*calipers*).

pregnancy are benign cystic teratomas. These can usually be identified by their sonographic characteristics (eFig. 9.29). Ovarian malignancy is rare, so arbitrary size cutoffs to dictate surgery are not usually used. Consultation with a gynecologic oncologist may be prudent when a suspicious mass is seen. If an adnexal mass is suspected based on transabdominal second- and third-trimester scans, a transvaginal approach should yield more useful information because the probe is closer to the mass in question and higher-frequency ultrasound can be utilized for better definition.

Standard Measurements

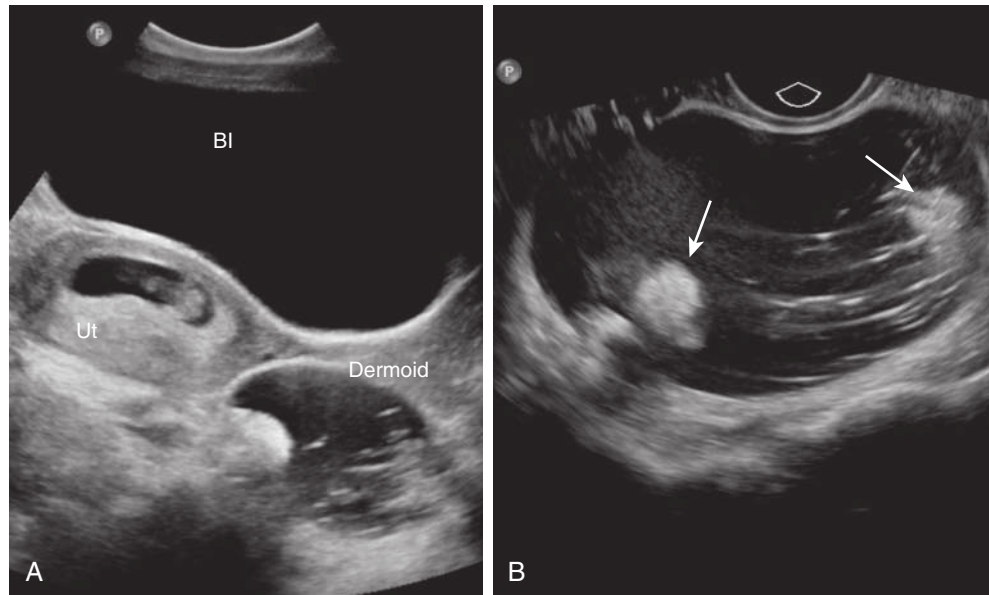
With transvaginal ultrasound, the fetal pole can be measured when it is only a few millimeters long, between 5 and 6 weeks of gestation. From 6 to



Fig. 9.25 Measurement of the Crown-Rump Length (*CRL*). Sagittal view of the fetus with the crown of the head to the right and the rump to the left.

10 weeks of gestation, the maximal length of the embryo is measured. After about 10 weeks, the head, trunk, and extremities are visible, and the measurement is taken from the well-visualized crown to the rump (Fig. 9.25 and eFig. 9.30). The CRL becomes less accurate after the first trimester and should not be used after 13 weeks of gestation.⁵⁰

It is possible to measure virtually any body part of the fetus. However, a standard set of measurements has been accepted as being the most useful for gestational age prediction. After the first trimester, these include the biparietal diameter (BPD), head circumference (HC), abdominal circumference (AC), and femur length (FL). Unless these are measured according to standard criteria (Box 9.6), their value in



eFig. 9.29 Transabdominal and Transvaginal Images of an Ovarian Dermoid. (A) An 8-week pregnancy with the uterus (*Ut*) and bladder (*Bl*) demarcated in the transabdominal image. (B) Using transvaginal ultrasound, the mass fills the entire field of view. Characteristic irregular mixed solid and cystic areas with highly echogenic nodules (*arrows*) are seen within the dermoid. To see the pelvic structures clearly with transabdominal ultrasound, the maternal bladder must be very full. Although this provides a good window, the structures of interest are pushed away from the probe, which limits resolution. In contrast, the probe is less than 1 cm from the mass when transvaginal ultrasound is used.



eFig. 9.30 This is an inappropriate image for obtaining the crown-rump length (*CRL*) because the flexed position of the fetus gives a falsely low measurement.

BOX 9.6 Description of Images for Fetal Biometry

Head

- Image is from side, not front or back.
- Head is oval rather than round.
- Midline structures are centered, not displaced to the side.
- Measure at the level of thalamus and cavum septum pellucidum.
- Image should not include the top of the orbits or any part of the brainstem or cerebellum.
- Measure outer edge of the proximal skull to the inner edge of the distal skull.
- Head circumference is measured at the level of the biparietal diameter, around the outer perimeter of the skull.

Abdomen

- Abdomen is nearly round, not oval or squashed.
- Obtain a true transverse image, not oblique.
- Images of ribs should be similar on both sides.
- Measure at the level where the umbilical vein joins the portal sinus.
- Calipers are all the way to the skin surface, not at the rib, liver, or spine edge.

Femur

- Femur should be perpendicular to direction of insonation.
- Ends should be sharply visible, not tapered or fuzzy.
- The measurement should exclude the distal epiphysis.

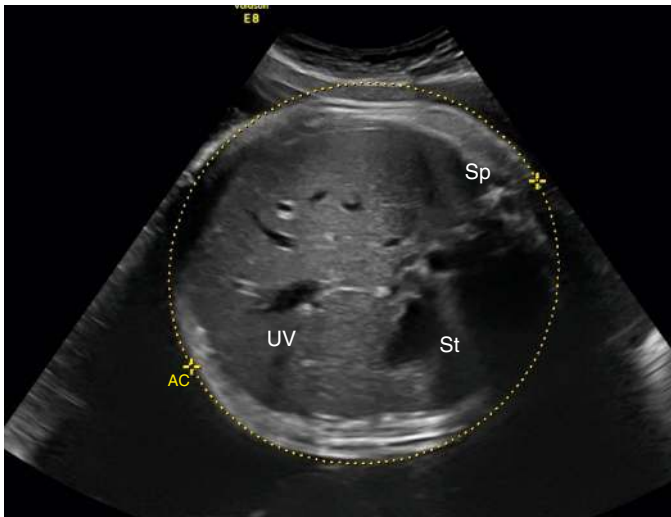


Fig. 9.26 Proper Image for Measurement of the Abdominal Circumference (AC). Criteria for an appropriate image are listed in [Box 9.6](#). The umbilical vein (UV) is seen at the level of the portal sinus. Sp, Spine; St, stomach.

defining the age or growth of a fetus is limited. Optimal views for obtaining measurements are shown in [Figs. 9.26–9.28](#). Common errors to be avoided are shown in [eFigs. 9.31 and 9.32](#). With practice, a sonographer and sonologist learns to slide the transducer to the proper position on the abdominal wall, then angulate and rotate the transducer until the precise plane for biometry or anatomic assessment is obtained.

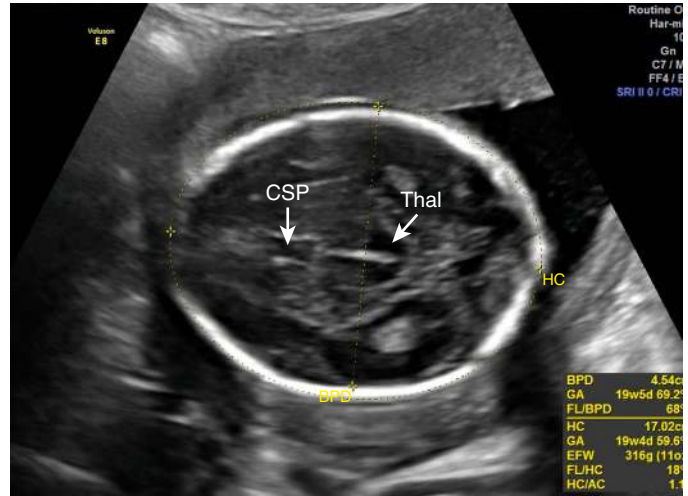


Fig. 9.27 Image of the Fetal Head for Biparietal Diameter (BPD) and Head Circumference (HC) Measurements. The cavum septum pellucidum (CSP) and the thalami (Thal) are indicated. Criteria for an appropriate image are listed in [Box 9.6](#).

Fetal Anatomic Survey

Systematic evaluation of fetal anatomy is critical. Although many well-recognized risk factors in the pregnant patient exist for congenital anomalies, most birth defects occur in the fetuses of low-risk people. For this reason, it is important that anyone who performs obstetric ultrasound has familiarity with the appearance of appropriate fetal anatomy in order to recognize potential deviations. As mentioned previously, although more advanced detailed sonograms (specialized examinations) are appropriate for patients with identified risk factors, **all standard examinations should include a full fetal anatomic survey.**^{6,35}

As previously noted, specific components should be evaluated during a standard ultrasound exam performed in the second or third trimesters (see [Boxes 9.4 and 9.5](#)). Typically, a single complete anatomic survey is recommended at 18 to 22 weeks of gestation, and a limited anatomic survey can also be performed when follow-up studies are requested. When components of the standard second- or third-trimester ultrasound examination cannot be obtained because of obesity, unfavorable fetal position, or other technical factors, this should be documented in the report. In such patients, it is reasonable to repeat the ultrasound in 2 to 4 weeks.

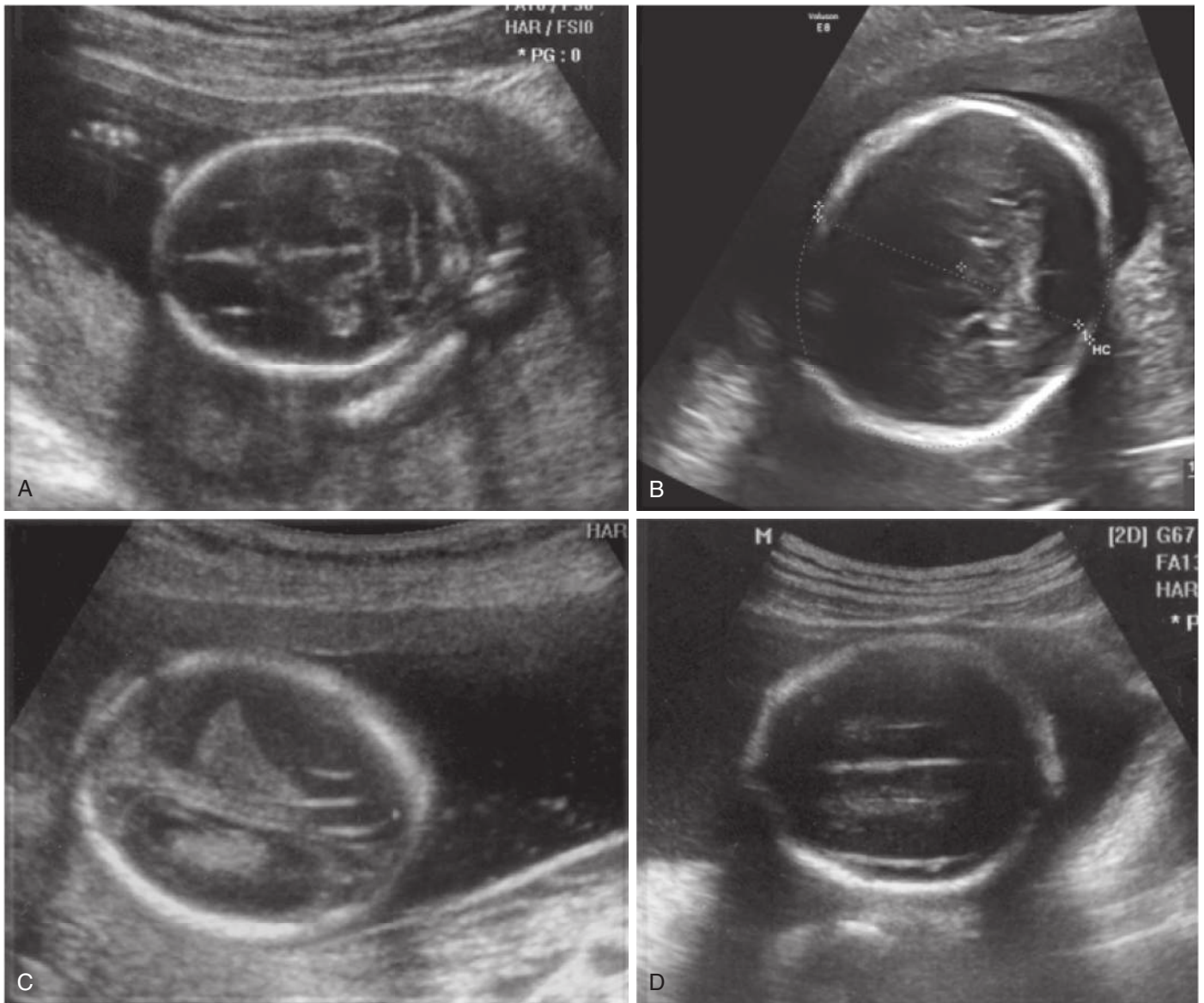
In the following sections, normal structures typically evaluated during an anatomic survey are discussed.

Documentation

The report of all ultrasound exams should list the indication. Documentation of ultrasound findings is important not only for good patient care but also for quality review and legal defense. AIUM and ACOG guidelines both emphasize that **adequate documentation of an ultrasound study is essential for high-quality patient care.**^{6,35} Images of all appropriate areas, both normal and abnormal, should be recorded. An official interpretation of the ultrasound findings should be included in the patient's medical record.

Intracranial Structures

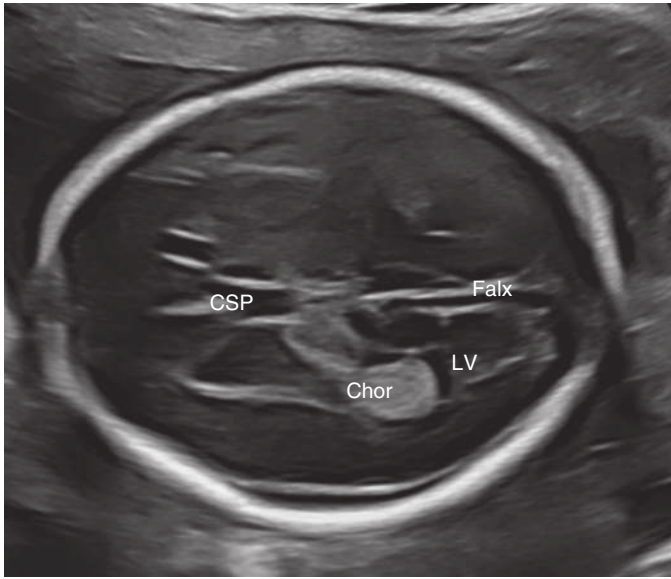
Screening views for fetal neurosonography typically involve axial views on the transventricular plane ([eFig. 9.33](#)), transthalamic plane ([Fig. 9.29](#)), and transcerebellar plane ([eFig. 9.34](#)).⁵¹ In these screening views, several structures will need to be assessed, including the



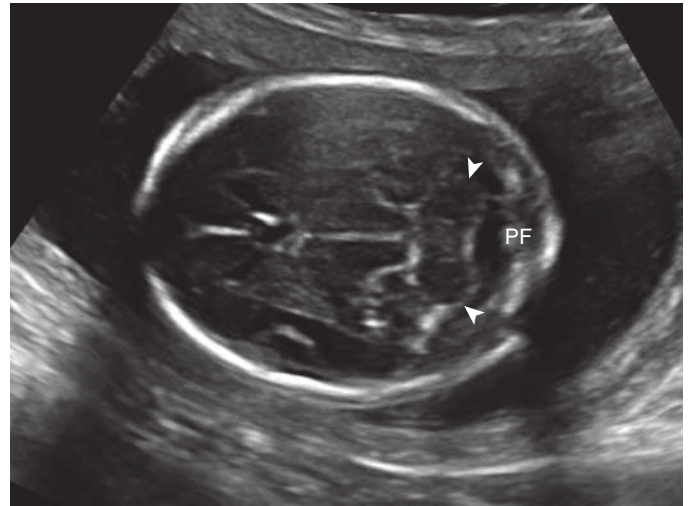
eFig. 9.31 Examples of Inappropriate Head Views for Biparietal Diameter and Head Circumference (*HC*) Measurements. (A) The transducer is rotated in such a way that the image includes the cerebellum. (B) The head is imaged from the back and not from the side. (C) The scan plane goes through the head obliquely. Note how the brain structures are asymmetric. (D) This true axial image is too high on the head. The cavum septum pellucidum is not seen.



eFig. 9.32 Examples of Abdominal Views That Are Inappropriate for Abdominal Circumference Measurements. (A) This true axial image is too low on the abdomen. The umbilical vein is seen close to the anterior abdominal wall. (B) This is an oblique cut through the abdomen. Note the asymmetric appearance of the ribs. (C) The abdomen is overly compressed because the sonographer has applied too much pressure to the maternal abdominal wall. (D) The transducer is rotated in such a way that an oblong view of the abdomen is obtained. This gives an artificially elongated anterior-posterior dimension.



eFig. 9.33 Axial View of the Fetal Cranium at the Level of the Lateral Ventricles. *Chor*, Choroid; *CSP*, cavum septum pellucidum; *Falx*, falx cerebri; *LV*, lateral ventricle.



eFig. 9.34 Normal Cerebellum. The two cerebellar hemispheres are indicated by the *arrowheads*. The cisterna magna is the fluid-filled space posterior to the cerebellum. *PF*, Posterior fossa.

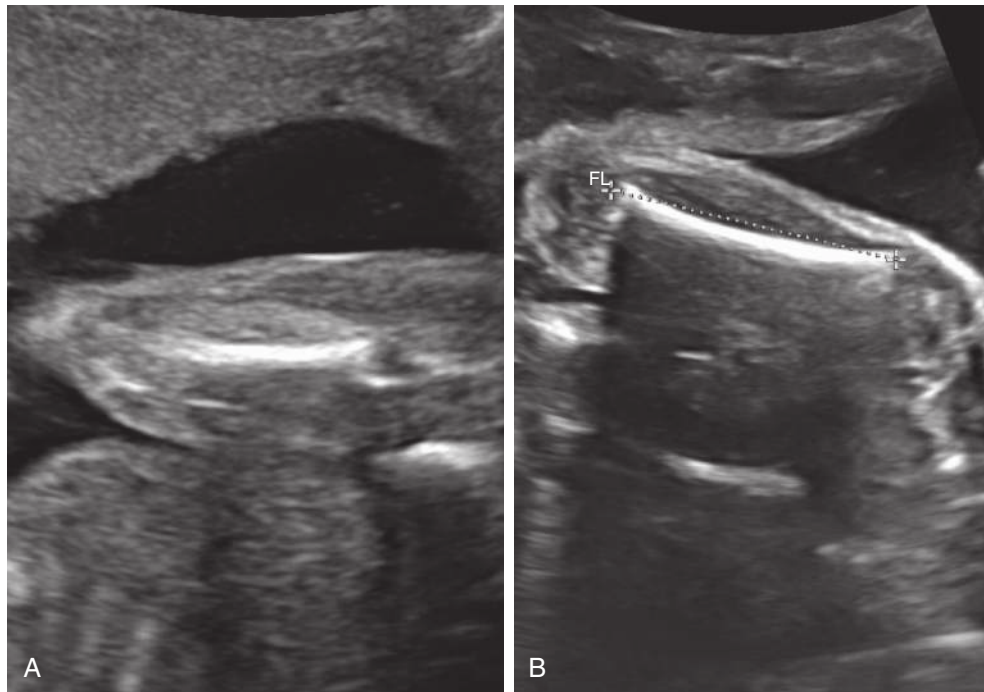


Fig. 9.28 (A) The entire diaphysis of the femur is not clearly shown. This will give an inappropriately short measurement. The ends should be distinct, as shown in B. *FL*, Femur length.

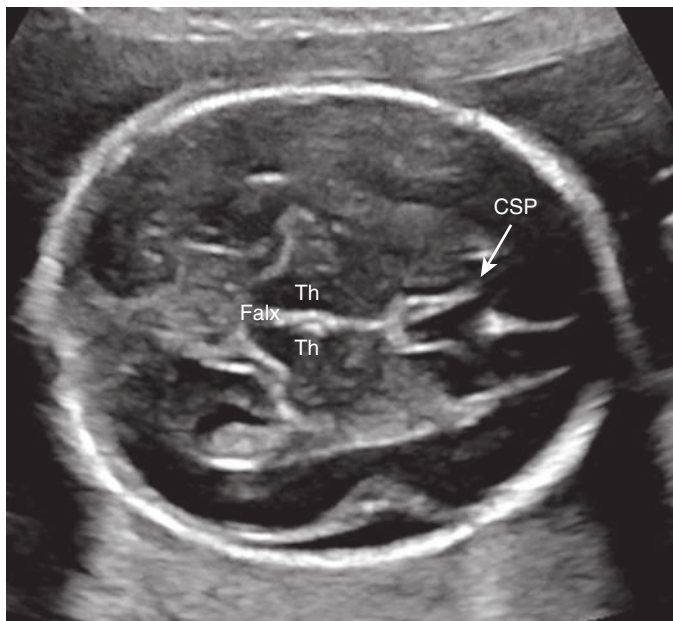


Fig. 9.29 Axial View of the Fetal Cranium at the Level of the Thalami. The cavum septum pellucidum (*CSP*), thalami (*Th*), and falx cerebri (*Falx*) can be evaluated in this view.

cranium shape, lateral ventricles, choroid plexus, cavum septi pellucidi, thalami, cerebellum, and cisterna magna. If an anomaly is suspected based on screening views or medical history, additional views, such as coronal, sagittal, and parasagittal views, may be used to better visualize intracranial structures.⁵²

Facial Structures

A coronal view of the fetal face is usually readily obtainable (eFig. 9.35). In this view, normal orbits and lenses can be seen. A midsagittal view of

the fetal face, also known as the profile view, is also important to obtain. In this view, a normal forehead, nasal bone, palate, and mandible can be assessed (eFig. 9.36). Additional axial views at the level of the palate and mandible may be helpful in ascertaining the integrity and the size of these structures.

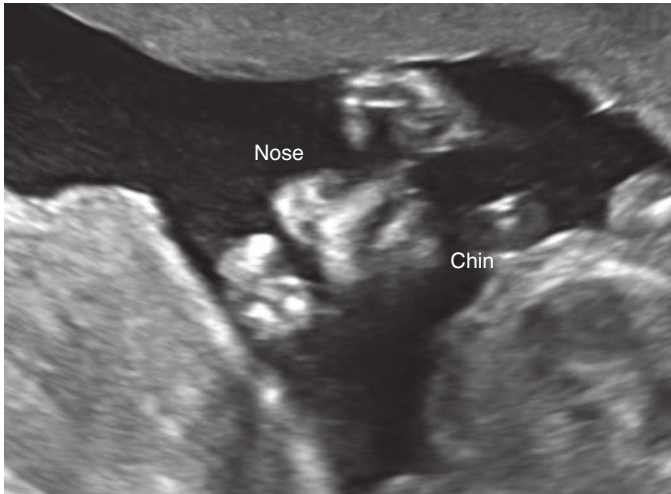
Spine

Normal transverse views of the spine show intact skin with a consistent relationship between the echocenters at each level (eFig. 9.37). Longitudinal views show the lateral echocenters running parallel to each other (eFig. 9.38). Both views are valuable in diagnosing open neural tube defects or sacrococcygeal teratoma.

Heart and Thorax

In addition to the position of the heart within the chest, the axis should be carefully noted (Fig. 9.30). The sonographer must identify the fetal right and left side based on the position of the fetus, not the position of the organs. Otherwise, abnormalities of sidedness—such as situs inversus—would be missed.

Standard views of the fetal heart typically include the four-chamber view (Fig. 9.31) and the left ventricular (eFig. 9.39) and right ventricular outflow tracts (eFig. 9.40). A good four-chamber view can allow for the assessment of the relative size and shape of all cardiac chambers and the atrioventricular valves (mitral and tricuspid valves). It is important to note that the fetal heart is a dynamic structure; as such, a cine-clip of the fetal heart cycle can often better demonstrate abnormalities of the fetal heart. Additional valuable axial views that can help screen for abnormalities involving the outflow tracts are the three-vessel and three-vessel trachea views (eFigs. 9.41 and 9.42). In a normal three-vessel view, the pulmonary artery and ascending aorta are of approximately equal size. In the three-vessel trachea view, the fusion of the ductal and aortic arches can be seen forming an acute angle. Doppler interrogation can also clarify the flow direction in the ductal and aortic arches. Aberrant vessels such as a persistent left superior vena cava can also be seen in this view.



eFig. 9.35 Coronal View of the Lower Fetal Face Showing a Normal Mouth.



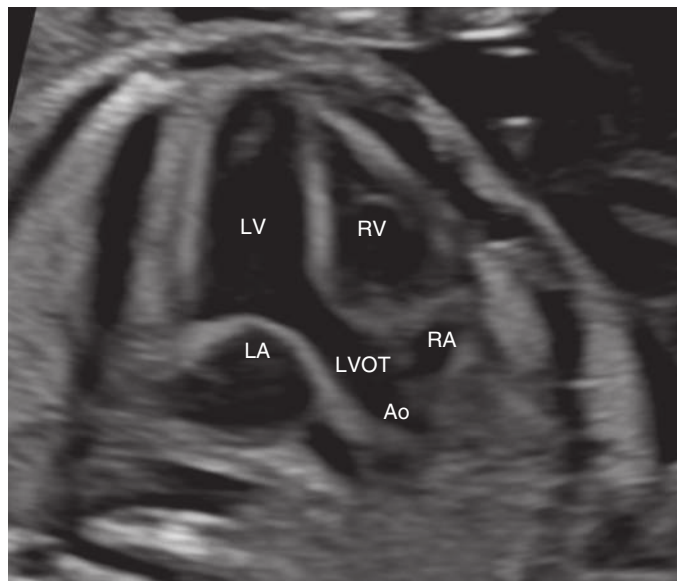
eFig. 9.37 Normal Transverse Abdominal View of a 20-Week Fetus. Notice the relationship between the echocenters of the spine (*arrowheads*), and the intact skin.



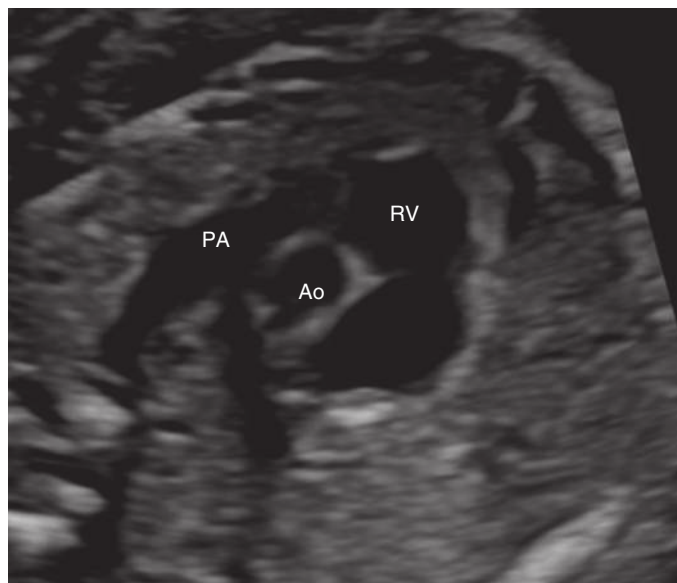
eFig. 9.36 Sagittal View of a Fetal Face Showing Micrognathia. The small mandible is indicated by the *arrowhead*.



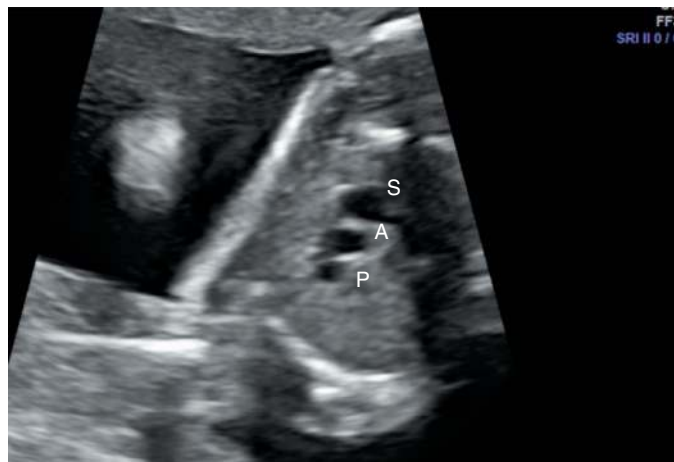
eFig. 9.38 Normal Sagittal View of the Fetal Spine.



eFig. 9.39 Left Ventricular Outflow Tract View, Also Known as the Five-Chamber View. This view demonstrates the path from the left ventricle into the ascending aorta (*Ao*). *LA*, Left atrium; *LV*, left ventricle; *LVOT*, left ventricular outflow tract; *RA*, right atrium; *RV*, right ventricle.



eFig. 9.40 Right Ventricular Outflow Tract View. This view demonstrates the path from the right ventricle to the main pulmonary artery (*PA*). Notice the branching of the pulmonary artery, which needs to be visualized for this view. *Ao*, Aorta; *RV*, right ventricle.



eFig. 9.41 Axial Cut Above the Level of the Heart Showing the Three-Vessel View. The superior vena cava (*S*), ascending aorta (*A*), and pulmonary artery (*P*) are approximately equal in diameter.



eFig. 9.42 Three-Vessel Trachea View. This view demonstrates the pulmonary artery (*PA*) and ductus arteriosus forming a V-shape connection with the aorta (*Ao*). This connection should run to the left of the trachea (*T*), confirming the left-sidedness of the arches. *S*, Superior vena cava.

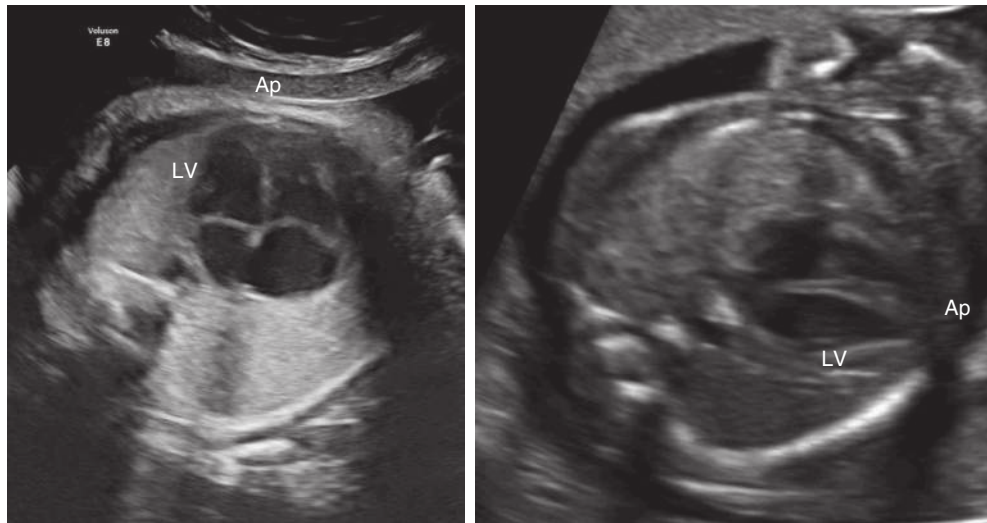


Fig. 9.30 Normal Four-Chamber Views. The image at *left* shows a fetus in a cephalic presentation, with the fetal left toward the upper part of the screen. The image at *right* shows a fetus in a breech presentation. Note that the heart's position is central in the chest and the axis is about 45 degrees to the left of the midline. The left ventricle (LV) and apex (Ap) of each heart is annotated.



Fig. 9.31 Four-Chamber Heart View, One of the Standard Screening Views of the Fetal Heart. LA, Left atrium; LV, left ventricle; PV, pulmonary vein; RA, right atrium; RV, right ventricle.

Common sagittal views of the fetal heart include aortic ductal arches. The aortic arch, which can be recognized by its “candy cane” shape (eFig. 9.43) and the presence of head and neck arteries, arises from the left ventricle and runs anterior to the main pulmonary artery, which arises posteriorly from the right ventricle. The ductal arch, on the other hand, has a distinctive “hockey stick” appearance (eFig. 9.44).

Aside from the fetal heart, the echotexture of the lungs and the presence of aberrant structures inside the chest, such as bowel or liver, should be noted. Right (eFig. 9.45) and left (eFig. 9.46) parasagittal views of the abdomen and thorax can be used to visualize the integrity of the diaphragms. The presence of aberrant structures in the fetal chest can often result in the deviation of the cardiac axis, which can prompt a more thorough fetal chest evaluation.

Abdomen

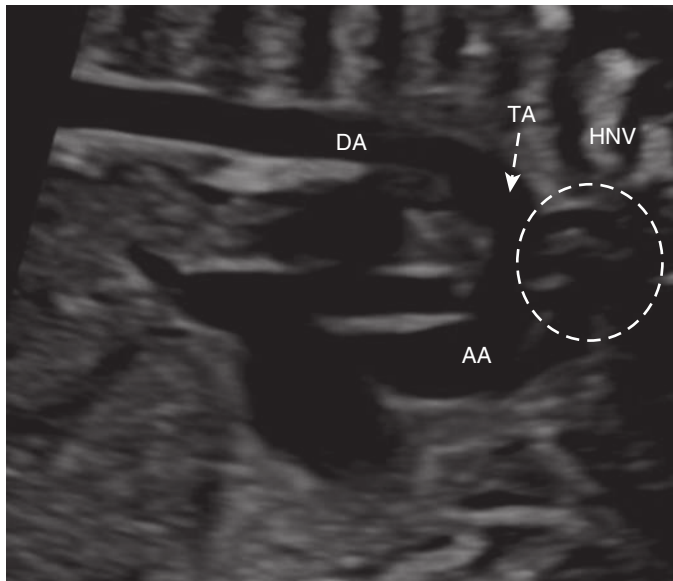
The fetal stomach should always be visualized after the first trimester. It is important to ascertain the position of the stomach to help evaluate for laterality defects. An axial view at the level of the stomach will also allow for the visualization of the liver, umbilical vein, and portal sinus (see Fig 9.26). The integrity of the abdominal wall can be demonstrated with the axial view at the level of the abdominal cord insertion (eFig. 9.47).

Urinary Tract

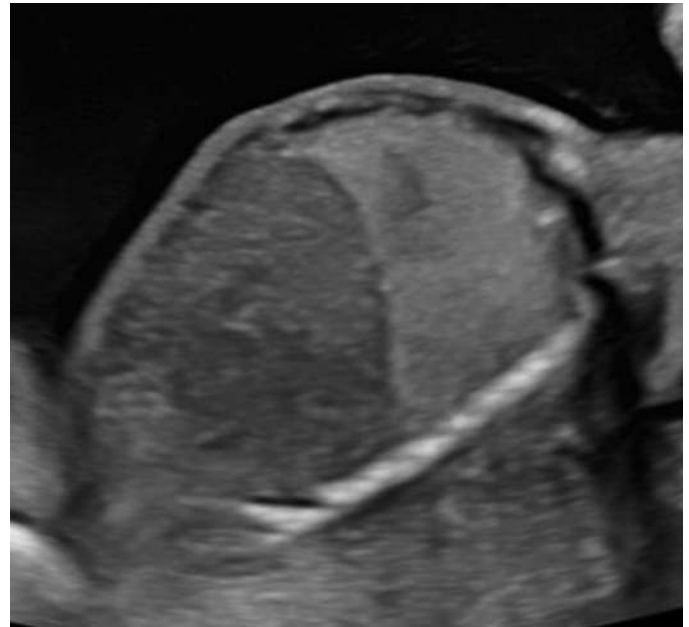
The evaluation of the fetal urinary tract typically includes the kidneys and bladder. Kidneys are commonly assessed in the axial view, with the spine at a 12 o'clock or 6 o'clock position (eFig. 9.48). This view is also used to measure the renal pelvis in the case of urinary tract dilation. The size and shape of the bladder can be evaluated in the coronal view. The addition of color Doppler in this view will also allow for the assessment of the quantity of the umbilical arteries. In certain cases, the coronal view of the kidneys may be needed, and the addition of color Doppler in this view can demonstrate the presence of both renal arteries (Fig. 9.32).

Skeletal

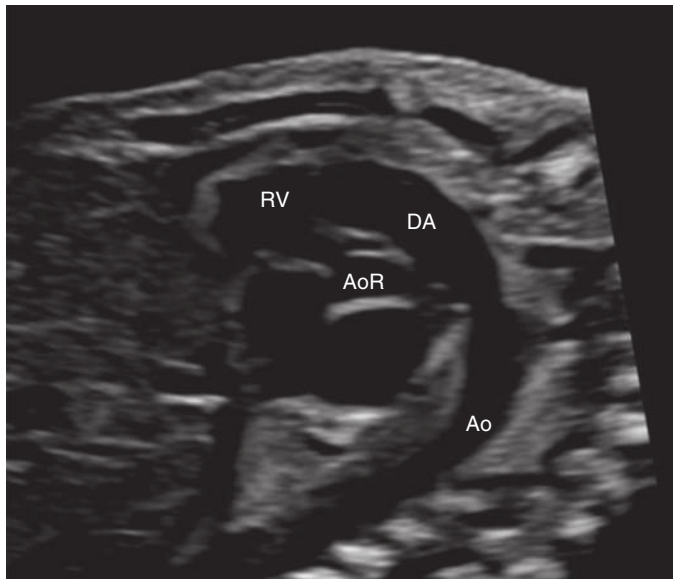
Upper and lower extremities need to be visualized in an anatomic survey. All the long bones need to be evaluated for the presence of morphologic abnormalities (fracture, bowing, or shortening) and their echogenicity. These qualities can be altered in cases of skeletal abnormalities, such as skeletal dysplasia. The hands and feet need to be evaluated for their alignment and appearance. Common abnormalities that can be seen include arthrogryposis, clubfoot, and poly- or syndactyly.



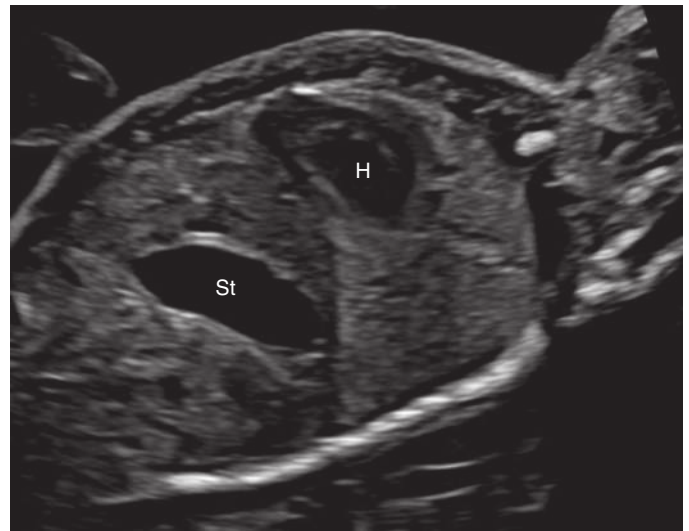
eFig. 9.43 Sagittal view of the aortic arch, demonstrating the course of the aorta starting at the ascending aorta (AA), transverse arch (TA), and descending aorta (DA). Head and neck vessels (HNV) can be seen branching out from the transverse arch. Notice the “candy cane” shape of the aortic arch.



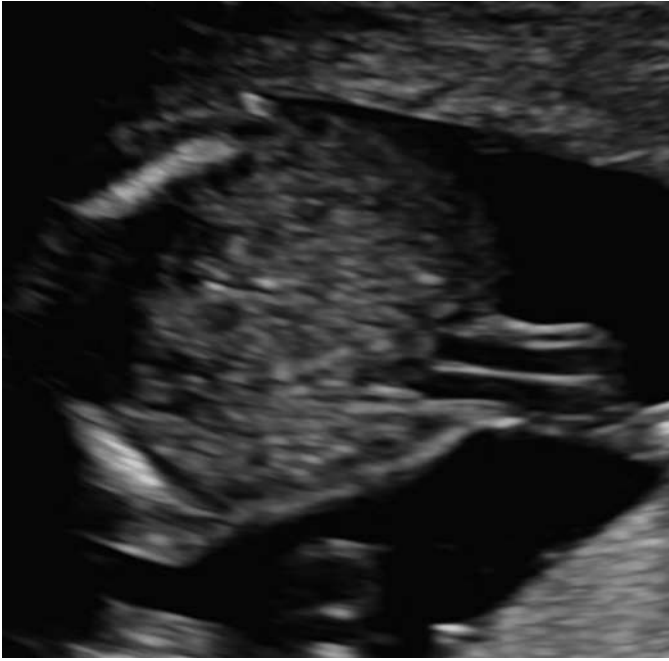
eFig. 9.45 Right parasagittal view of the abdomen and thorax, demonstrating intact right diaphragm separating the liver from the lung parenchyma.



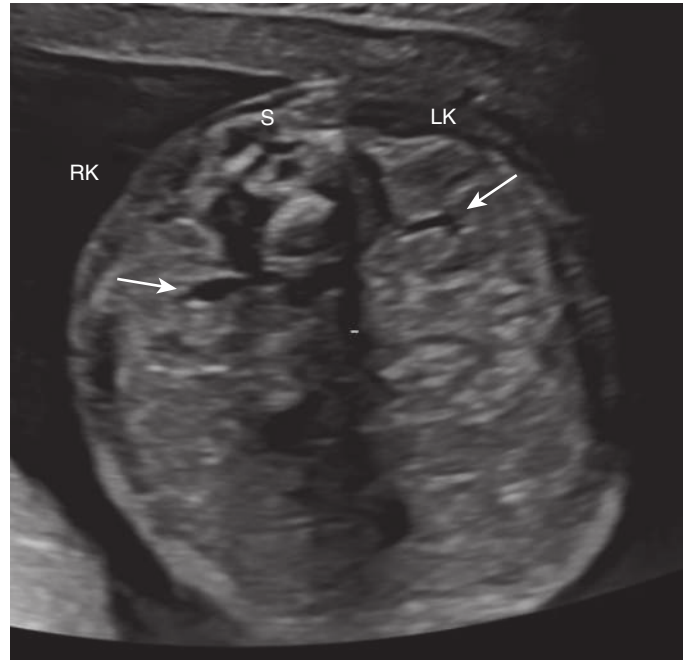
eFig. 9.44 Sagittal view of the ductal arch (DA), demonstrating the course of the ductus arteriosus toward the aorta (Ao), forming a distinctive “hockey stick” shape. The right ventricle (RV) and aortic root (AoR) can also be seen in this view.



eFig. 9.46 Left parasagittal view of the abdomen and thorax, demonstrating intact left diaphragm separating the stomach (St) and the heart (H).



eFig. 9.47 Axial cut of the fetal abdomen at the level of the abdominal cord insertion. This view demonstrates the integrity of the abdominal wall in this level.



eFig. 9.48 Axial view of the fetal abdomen with the spine (*S*) at around 12 o'clock position, demonstrating the presence of the right (*RK*) and the left (*LK*) kidneys. Bilateral renal pelvices need to be visualized in this view (*arrows*).

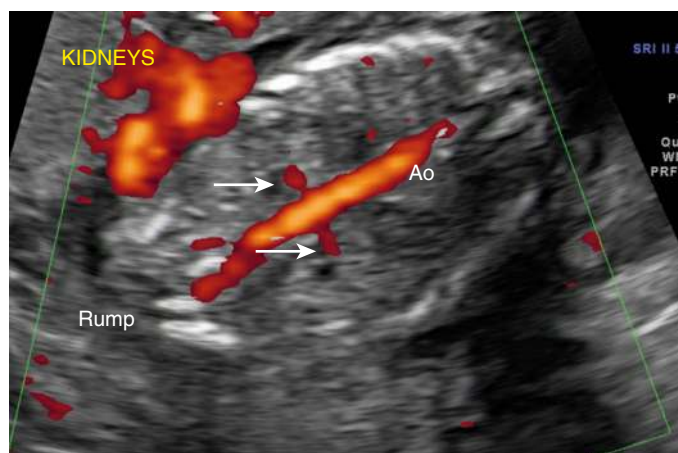


Fig. 9.32 Coronal Image Showing the Renal Arteries (Arrows) Arising From the Descending Aorta (Ao). This view helps confirm the presence of kidneys at a gestational age when the outlines of the kidneys may be indistinct.

CLINICAL APPLICATIONS OF ULTRASOUND

Gestational Age Determination

In very early pregnancy, the gestational sac diameter can give a close approximation of the gestational age. Because the range of error in age prediction from sac measurement is significantly greater than for the CRL and other standard measurements, the sac size should not be used as the final gestational age determinant. **The measurement of the early embryo, either as the maximal embryo length or CRL, is commonly accepted as the best sonographic method for gestational age determination.**^{6,50} Using a static scanner in pregnancies dated by “good” menstrual dates, Robinson and Fleming found that the 95% confidence interval for gestational age prediction using the CRL was ± 4.7 days.⁵³ Later studies showed an even smaller random error when gestational age was determined by in vitro fertilization.⁵⁴ The high accuracy of ultrasound dating of early embryos is explained by their relatively large weekly percentage change in size and the fact that very little variation is seen among early embryos in their growth rates. **The variability of gestational age estimations increases with advancing gestation.**

The 95% confidence interval for gestational age prediction using the BPD as a single parameter between 14 and 21 weeks of gestation is ± 7 days.⁵⁵ In a study of pregnancies dated by in vitro fertilization, Chervenak and colleagues found that in this period of pregnancy, all of the single parameters considered (BPD, HC, and FL) performed quite well: 95% confidence intervals for the gestational age ranged between ± 7.5 days for the HC and ± 8.7 days for the FL.⁵⁶ Composite gestational age assessment using mathematic modeling from multiple parameters is now the most accepted method of ultrasound dating after the first trimester.⁵⁰

When to Use Ultrasound Dating

Since the time of Nägele in the early 1800s, obstetric providers have routinely used menstrual dates for determining the estimated date of delivery (EDD). Agreement is near universal that in pregnancies in which the menstrual dates are uncertain or thought to be unreliable

TABLE 9.1 Circumstance in Which Ultrasound Gestational Age Determination Should Take Precedence Over “Good” Menstrual Dates^{3,20,47}

Gestational Age Range ^a	Method of Measurement	Discrepancy Between Ultrasound Dating and LMP Dating That Supports Redating
<8 ^{6/7}	CRL	>5 days
9 ^{0/7} to 13 ^{6/7}	CRL	>7 days
14 ^{0/7} to 15 ^{6/7}	BPD, HC, AC, FL	>7 days
16 ^{0/7} to 21 ^{6/7}	BPD, HC, AC, FL	>10 days
22 ^{0/7} to 27 ^{6/7}	BPD, HC, AC, FL	>14 days
$\geq 28^{0/7b}$	BPD, HC, AC, FL	>21 days

^aBased on LMP.

^bBecause of the risk of redating a small fetus that may be growth restricted, management decisions based on third-trimester ultrasonography alone are especially problematic and need to be guided by careful consideration of the entire clinical picture and close surveillance.

AC, Abdominal circumference; BPD, biparietal diameter; CRL, crown-rump length; FL, femur length; HC, head circumference; LMP, last menstrual period.

From Committee Opinion No. 700: Methods for estimating the due date. *Obstet Gynecol.* 2017;129:e150-e154.

because of uncertain recall, irregular cycles, recent hormonal contraception use, or an abnormally light last menstrual period (LMP), dates derived from ultrasound biometry are preferred.^{6,50} A joint committee opinion issued in 2017 from the ACOG, the AIUM, and the SMFM titled “Methods for Estimating the Due Date” made very specific recommendations for ultrasound dating that extend through the entire pregnancy (Table 9.1).⁵⁰ Importantly, the committee opinion and the 2016 ACOG practice bulletin both note that **in the third trimester, aberrant growth is relatively common, so changing certain menstrual dates in favor of a late ultrasound date risks missing the presence of a significant growth disorder, and decisions about changing dates require careful consideration and close surveillance.**^{6,50}

When the first ultrasound is performed after 22 weeks 0 days, the pregnancy is considered “suboptimally dated.”⁵⁷ In spite of the pitfalls of dating a pregnancy based on a late ultrasound, this may be the only option available. The ACOG has provided guidelines on how suboptimally dated pregnancies should be managed. One step is to perform a subsequent ultrasound 3 to 4 weeks after the initial dating scan. An abnormal growth curve may be a clue that the assigned EDD is incorrect.

Using Ultrasound to Assess Fetal Growth

The EDD determined at the time of the first ultrasound examination forms the basis for subsequent clinical and sonographic assessments of fetal growth. Because insufficient or excessive growth can be

associated with significant fetal morbidity, recognition of growth abnormalities is one of the primary aims of prenatal care and is an important aspect of ultrasound examinations. In most obstetric practices, growth ultrasound examinations are obtained when abdominal palpation and fundal height measurements raise the suspicion of abnormal growth.⁵⁸ A lagging fundal height, combined with the knowledge of clinical risk factors for FGR, is commonly used as a screening test for insufficient fetal growth, whereas ultrasound is considered the diagnostic test. This scheme for the prenatal detection of growth-restricted fetuses has been met with mixed results. Although some studies have shown sensitivities for the detection of small-for-gestational-age (SGA) fetuses with lagging fundal height to be as low as 15%,⁵⁸ others suggest a 65% to 85% sensitivity and 96% specificity.⁵⁹ The positive predictive value of a lagging fundal height for diagnosing FGR has been reported to be 50%. Schemes for the antenatal diagnosis of small fetuses may perform significantly better if clinical risk factors are considered.

Serial ultrasound examinations have been shown to be superior to physical examination for diagnosing insufficient growth.⁶⁰ For this reason, some have advocated for a routine third-trimester ultrasound examination to screen for growth abnormalities even in low-risk pregnancies. Studies have shown, however, that this policy has little or no impact on perinatal morbidity and mortality.^{59,61} Because of the high cost and unproven benefit of such a screening program, routine ultrasound examinations for growth are generally not a part of prenatal care in the United States. However, it is still accepted that ultrasound should be ordered when clinical examination is suspicious and when significant risk factors for a growth abnormality are present. Such risk factors in the pregnant patient include diabetes mellitus, hypertension, a previous small or large baby, multiple gestation, or poor weight gain.⁵⁹ A risk-based approach is particularly appropriate in cases such as multiple gestation or marked obesity of the pregnant patient, in which clinical evaluation of fetal size is difficult or impossible.

Estimating Fetal Weight

Several formulas for calculating the estimated fetal weight (EFW) are in use.⁶² Recent attempts at incorporating a larger and more diverse cohort to create more accurate formulas for EFW have been made.^{63,64} All incorporate the AC because this is the standard measurement most susceptible to variations in fetal soft tissue mass. Although the AC alone is a fairly good marker of abnormal fetal growth, the addition of other standard measurements to estimated weight formulas increases accuracy.⁶⁵ It has been shown that the addition of measurements beyond the standard set (BPD, HC, AC, and FL) does not significantly improve weight estimations. It appears that the error inherent in obtaining the basic measurements, especially the AC, is great enough to obscure any refinement in accuracy that might be gained from additional measurements.

Many studies have been done to evaluate the accuracy of ultrasound for predicting birthweight.⁶²⁻⁶⁴ Ultrasound accuracy in the prediction of fetal weight appears to have improved in recent years. However, all formulas, including newer formulas incorporating a more diverse cohort, appear to produce some degree of random error.⁶⁶ While some studies suggest the superiority of the Hadlock formula, the differences in accuracy across formulas are usually of limited clinical significance, and no formula has been consistently shown to be superior.^{63,64,66,67}

When scans are done by well-trained sonographers, most studies show a mean absolute error in weight prediction of about 8% to 10% when compared to the actual birthweight. Overall, about three-

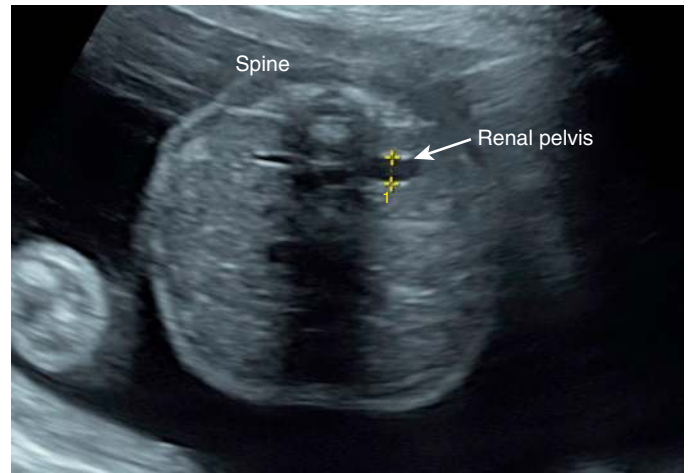


Fig. 9.33 Midabdominal Axial View Showing Urinary Tract Dilation Grade A1. The anterior-posterior measurement of the left renal pelvis (calipers) is shown. The spine is oriented at the 12 o'clock position.

quarters of infants have a birthweight within 10% of ultrasound estimates.⁶⁸ In 5% of cases, the error is greater than 20%.⁶⁹

Screening for Aneuploidy

Well-established screening tests exist for fetal aneuploidy; however, the role of ultrasound in aneuploidy screening is less straightforward. Starting in the 1980s, a number of soft markers (as opposed to actual birth defects) were found to have an association with Down syndrome. These include a thick nuchal fold (eFig. 9.49), urinary tract dilation (Fig. 9.33), echogenic bowel (eFig. 9.50), hypoplastic nasal bone (eFig. 9.51), cerebral ventriculomegaly (eFig. 9.52), short femur or humerus, and an echogenic intracardiac focus (eFig. 9.53). It has been shown that these markers perform poorly as individual predictors of Down syndrome.⁷⁰ **In spite of the well-documented association between these markers and trisomy 21, the ultrasound at 18 to 20 weeks of gestation should not be considered an adequate screening test.** A patient who desires such screening should be encouraged to use better-defined and well-established methods, which may include first-trimester screening, the quad screen, cell-free fetal DNA testing, or diagnostic testing.

Not infrequently, soft markers are seen incidentally at the time of the screening exam at 20 weeks. The ACOG has made recommendations on how to proceed in this circumstance.⁷⁰ In the setting of a low-risk aneuploidy screening result, the presence of isolated ultrasound soft markers, such as echogenic intracardiac focus or choroid plexus cyst, does not confer significant risk for aneuploidy. On the other hand, detailed anatomic assessment in the first or second trimester is paramount after a high-risk screening modality result. This will include assessment of both structural abnormalities and the presence of soft markers. The result of these assessments may provide guidance in choosing the appropriate diagnostic modality should it be desired.

Accuracy of Ultrasound as a Screening Tool for Birth Defects

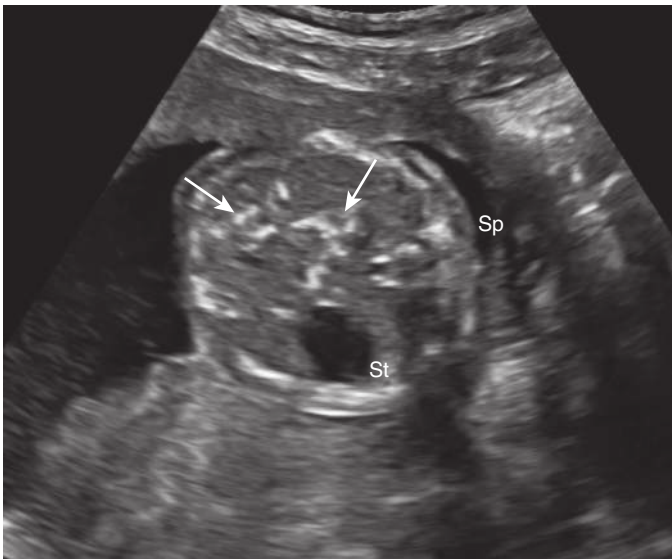
In the 1980s and early 1990s, several individual centers reported fetal anomaly detection rates of greater than 75% in both referral and low-risk patients. However, the sensitivity of routine ultrasound exams in more general settings has been found to be significantly lower.⁷¹⁻⁷⁴ In a large, randomized trial of routine ultrasound conducted in the 1990s,⁷¹



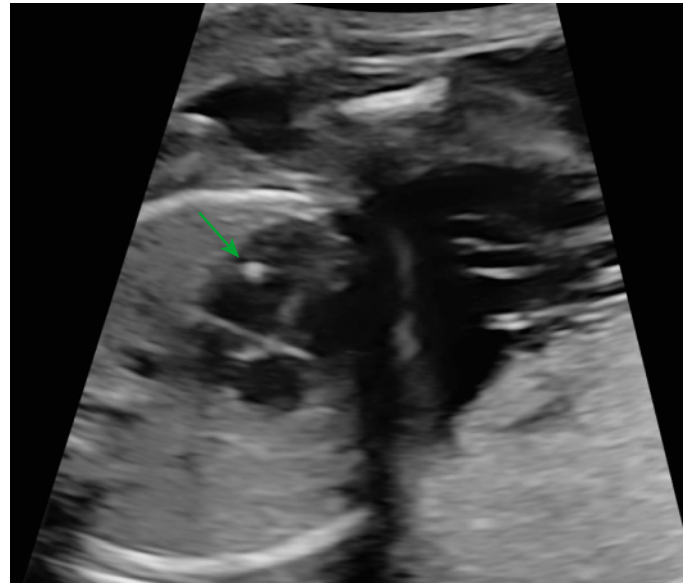
eFig. 9.49 Axial Image of a Fetus With Trisomy 21, Showing a Thick Posterior Nuchal Fold (*Calipers*).



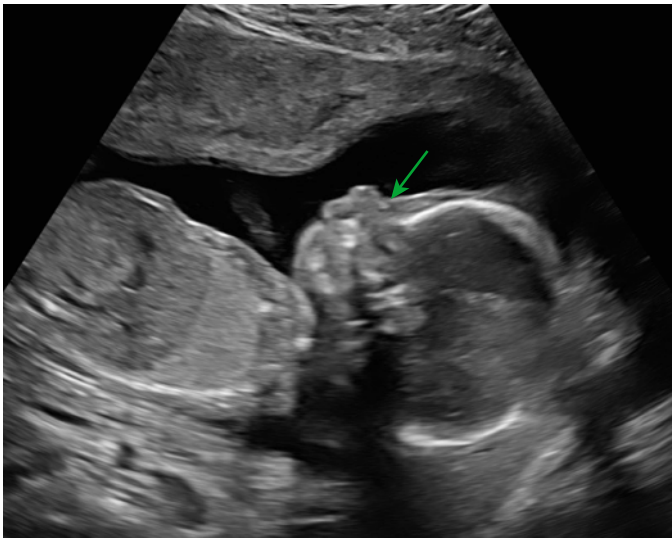
eFig. 9.52 Axial Cranial View of a 20-Week Fetus Showing Very Mild Ventriculomegaly (*Calipers*). The normal width at the atrium of the lateral ventricle is less than 10 mm.



eFig. 9.50 Midabdominal Axial View of Fetus With Trisomy 21 Showing Echogenic Bowel (*Arrows*). *Sp*, Spine; *St*, stomach.



eFig. 9.53 Axial View of the Fetal Heart Showing an Echogenic Intracardiac Focus in the Left Ventricle (*Green Arrow*).



eFig. 9.51 Midsagittal View of a 20-Week Fetus Showing a Hypoplastic Nasal Bone (*Green Arrow*).

BOX 9.7 Conditions for Which Magnetic Resonance Imaging May Be Helpful

- Spina bifida—Helps to document the level of the open defect, confirm the presence of a Chiari II malformation, and distinguish myelomeningocele from other lesions.
- Cervicofacial anomalies and tumors—Helps define the status of the fetal airway, which can be critical in guiding immediate neonatal care.
- Congenital diaphragmatic hernia—Confirms whether liver is in the chest, and can calculate the volume of lungs.
- Sacrococcygeal teratoma—Can help distinguish this lesion from terminal neural tube defects, determine any intraabdominal component of the tumor, and show evidence of compression of the spinal canal or other pelvic structures.
- Abdominal-pelvic masses—Helps confirm the location of masses (e.g., retroperitoneal), and can identify pelvic organs in suspected cloacal malformations.

defects of the central nervous system and urinary tract were detected 88% of the time, whereas only 18% of cases of cleft lip and palate were diagnosed prenatally. Ultrasound identified 74% of major musculoskeletal malformations and 39% of major cardiac defects. Of minor anomalies in the musculoskeletal and cardiac systems, 18% and 21%, respectively, were diagnosed prenatally. In addition, there are several conditions for which fetal magnetic resonance imaging may be useful as a supplement to ultrasound (Box 9.7).

In the hands of a well-trained examiner, the detection rate of some anomalies should approach 100%. For example, anencephaly, abdominal wall defects, and anomalies that involve an abnormal accumulation of fluid within a body cavity (e.g., urinary tract dilation or cerebral ventriculomegaly) will rarely be missed. The fact that spina bifida, which itself may be difficult to see, is almost always accompanied by easily recognized intracranial findings makes the detection rate of this defect quite high. Studies have shown that the detection of heart defects is significantly higher when views of the cardiac outflow tracts are obtained in addition to the four-chamber cardiac view.⁷⁵ The ACOG and the AIUM now recommend that these views be included in second- and third-trimester scans.^{6,35} On the other hand, congenital anomalies such as isolated palatal defect, clubfoot, or coarctation of the aorta may be harder to diagnose prenatally as they have been shown to have a significant false-positive rate.

The detection rate of birth defects has been shown to be higher in high-risk populations screened at centers with extensive experience at diagnosing birth defects. **Adequate training is essential for the proper performance and interpretation of obstetric ultrasound exams. To maintain acceptable levels of sensitivity, compliance with ACOG and AIUM guidelines for training and experience is important.**

The gestational age at which ultrasound is performed can influence the sensitivity of anomaly screening. Some defects may become more readily apparent late in pregnancy, while defects such as lissencephaly may not be detectable at an early gestational age. Although multiple exams in each pregnancy would undoubtedly increase the detection rate of malformations, resource and financial constraints make this impractical.

Summarizing the results of large studies and systematic reviews, the report of the 2014 Fetal Imaging Workshop hosted by the National Institute of Child Health and Human Development (NICHD) concluded that the range of detection rates for anomalies overall is wide, from 16% to 44% prior to 24 weeks of gestation, with a detection rate of 84% for major and lethal anomalies.³⁹

It is generally agreed that the most cost-effective approach for screening low-risk pregnancies is to perform a standard examination between 18 and 22 weeks of gestation. At this gestational age, adequate visualization of major organs is possible for most patients. In some, because of an unusually thick or dense abdominal wall in the pregnant patient or an unfavorable fetal position, completion of the anatomic survey is suboptimal, and a follow-up scan may be helpful. Another newer approach in anatomic surveys for pregnant people with increased adiposity is the use of transvaginal ultrasound in the late first trimester to the early second trimester. This approach may be able to improve the detection rate for congenital anomalies, although it is not yet a standard of care.⁷⁶⁻⁷⁸

Clinical Value of Prenatal Diagnosis of Birth Defects

Prenatal diagnosis allows the care team time to gather complete information about a fetal problem to best tailor management approaches to the needs of the family. In certain cases, options such as pregnancy termination or providing palliative or comfort care at birth can be explored. Prenatal consultation with pediatric subspecialists can also be facilitated to improve parental understanding and at the same time prepare the medical team for postnatal management. Prenatal diagnosis ensures that arrangements can be made for delivery at the proper time at a high-risk perinatal center where the newborn can receive optimal care.

Historically, options for intrauterine therapy have been limited. However, tremendous advancements have been made in recent years that have expanded this field; some available options for fetal therapy currently include intrauterine fetal transfusion, vesico- or thoracoamniotic shunt placements, fetal laser photocoagulation, fetoscopic endoluminal tracheal occlusion (FETO), and meningo-myelocele repair.⁷⁹ The options for fetal therapy will continue to grow with continuing progress in the development of fetal gene therapy.⁸⁰

Knowledge regarding anatomic diagnoses can also help clinicians select appropriate genetic testing. In the era of genomic medicine and with increasing use of broad-based testing, such as whole-exome sequencing, the importance of complete phenotyping using prenatal imaging cannot be stressed enough. With improved phenotyping, the likelihood of obtaining diagnostic results from whole-exome sequencing also increases.⁸¹ As of right now, the reported diagnostic yield of prenatal whole-exome sequencing varies greatly, but it appears to be less than the yield of the testing postnatally. As we continue to improve our ability to perform prenatal phenotyping, it is likely that the diagnostic yield will be similar to postnatal testing.⁸²

For many genetic conditions, an accurate diagnosis of the condition is important for calculating recurrence risks for future pregnancies and determining whether other family members are at risk. When a defect is detected prenatally, there is a greatly increased chance that the appropriate genetic tests or autopsy will be done.

KEY POINTS

- Sonographers should become familiar with the basic physics of ultrasound, equipment controls, safety of ultrasound, and scanning techniques to optimize ultrasound images.
- All of the elements of the standard obstetric ultrasound exam are important for clinical management and should not be neglected.
- Because most birth defects occur in the fetuses of low-risk patients, all standard exams should include a full anatomic survey.
- Physicians who perform and interpret obstetric ultrasound examinations should have appropriate training and experience.
- Pregnancy dating by ultrasound measurements is the most accurate early in pregnancy. Guidelines have been established for when ultrasound dates should be used over menstrual dates.
- Ultrasound for fetal weight estimation is of limited usefulness for the prevention of shoulder dystocia and other complications associated with fetal macrosomia.
- The AIUM and the ACOG discourage the nonmedical use of ultrasound for entertainment purposes.
- The sensitivity of ultrasound for the detection of birth defects depends on the level of training of examiners and the maintenance of a structured approach to the fetal examination.
- Individuals who perform prenatal sonography should be aware of the normal appearance of fetal structures.
- Appropriate documentation of ultrasound studies is important for good medical care.

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